The wage-amenities tradeoff: Do firms offset minimum wage increases by cutting job amenities?

by FERNANDEZ, Daniel *

The paper studies how firms respond to minimum wage increases. While previous studies focus on the employment-wage tradeoff, I develop a framework in which firms choose wages and job amenities, subject to binding wage floors. The model predicts that higher minimum wages compress the wage distribution but increase dispersion in job amenities. Furthermore, when firms must offer the same amenities to all employees, compensating differentials can propagate throughout the wage distribution. I test these predictions using matched employer-employee administrative data from Brazil. I recover firm-level job amenities using worker flows, and trace wage spillovers exploiting cross-state variation in the effective minimum wage. My findings support the model's predictions. Minimum wage increases reduced wage inequality while increasing dispersion in job amenities and wages above the statutory minimum. The approach helps reconcile the economic intuition that firms respond to higher wage floors with recent evidence of limited employment effects.

Latest version

What happens when governments increase minimum wages? Economists have long disagreed over whether wage floor policies actually benefit low-paid workers.[1] Whereas supporters of the policy emphasize earnings gains at the bottom of the wage distribution, critics warn of unintended consequences for employment among low earners. Despite the varied opinions, concerns about poverty and earnings inequality have made these policies a standard recommendation among international organizations and poverty experts, leading to their implementation worldwide.[2]

In Economics, the academic debate centers on a tension between economic intuition and recent empirical evidence regarding the employment-wage tradeoff. Canonical models predict that firms employing low-paid workers respond to a higher minimum wage by cutting labor, leading to layoffs and fewer vacancies (Brown et al., 1982). In contrast, empirical studies find no significant decline in employment following minimum wage increases (Addison et al., 2009; Dube et al., 2010; Addison et al., 2012; Allegretto et al., 2017; Cengiz et al., 2019). Critics of these findings, nonetheless, continue to dispute the results and question the approaches employed (Neumark et al., 2014; Clemens, 2019; Neumark, 2019).

In this paper, I focus on a feature often overlooked in the debate about the minimum wage: job amenities.[3] I argue that firms compensate for higher wage floors by reducing

* CEMFI, Calle Casado del Alisal 5, Madrid, Spain (e-mail: daniel.fernandez@cemfi.edu.es). I acknowledge financial support from Grant PRE2022-101381 funded by MCIN/AEI/10.13039/501100011033 and "ESF +", Fundación Carolina, and CEMFI. This paper has benefited from helpful comments of Jason Sockin, Joan Llull, Cauê Dobbin, Pascuel Plotkin, Tom Zohar, Javier Feinmann, Isaac Sorkin, Guillermo Caruana, and Juan Ivars, as well as participants at the BUDIE and FIRMS Workshop (CEMFI). All errors are my own.

[1] The debate over the effects of higher minimum wages dates back to Stigler (1946), who laid the foundations for a strong, though not universal, disdain for wage floor policies. Consensus began to break down in the early 1990s, with the work of David Card and Alan Krueger. Since then, the profession has witnessed numerous meta-analyses (e.g., Card and Krueger, 1995; Neumark and Wascher, 2008; Neumark, 2017; Allegretto and Reich, 2018; Clemens, 2019; Wolfson and Belman, 2019) and academic back-and-forth, such as Card and Krueger (1994, 2000) versus Neumark and Wascher (2000), or Neumark et al. (2014) versus Allegretto et al. (2017)

[2] Over the last two decades, wage inequality has declined in roughly two-thirds of countries worldwide. Nonetheless, wage gaps remain large, and some countries now experience a reversal of the decline.

[3] In previous studies, amenities are often used interchangeably with benefits or perks (Mas and Pallais, 2017).

job amenities among low-paid workers. By incorporating a wage-amenities tradeoff, my approach bridges the gap between the economic intuition that firms respond to minimum wage increases and the empirical findings of limited employment declines.

I propose a framework that explicitly depicts a tradeoff between wages and job amenities, yielding two intuitive empirical predictions. A higher minimum wage compresses the wage distribution; however, it worsens working conditions by reducing job amenities at low-paying firms.

Empirically, separating responses to minimum wages from unrelated shifts in firm compensation policies can be challenging. In fact, the model shows that if firms can freely adjust amenities across employees, then taking these predictions to the data would require comprehensive worker-level data. That is, all tangible and intangible non-wage attributes related to employment relationships.

The framework, nevertheless, offers an additional, testable implication when firms choose a single level of amenities for all employees. In such cases, firms that employ workers both at the minimum wage and above must reduce job amenities for everyone. As a result, highly paid workers require a compensating differential to remain indifferent, which ripples throughout the whole wage distribution.

Brazil provides a particularly suitable setting for testing the implications of the framework. First, the Brazilian government makes reasonable adjustments to the federal minimum wage each January. Compared to other settings where the statutory minimum remains fixed for long periods and changes abruptly, this feature addresses concerns in the literature that prior reflect short-run distortions rather than effective minimum wage increases (Card and Krueger, 2016; Clemens and Strain, 2021; Hurst et al., 2022). Second, I leverage institutional features of the Brazilian labor market that support the presumption that firms choose a single level of job amenities for all employees. Most notably, the consolidated labor laws and the widespread use of municipal-level collective agreements impose rigid limits on tailoring amenities to individual workers within the same establishment. Third, Brazil provides high-quality matched employer-employee data that are central to the empirical exercises in this study.

The two main empirical challenges the paper faces are recovering unobserved job amenities and tracing spillovers across the entire wage distribution in response to higher wage floors. For the first challenge, I use worker flows across firms and recover job amenities at the establishment level. Building on Sorkin (2018), I leverage the fact that job-to-job transitions initiated by the worker, namely quits, reveal preferences over firms. By incorporating information on wage changes across such transitions, following Lehmann (2025), I separate wage and non-wage attributes of compensation. For the second challenge, I replicate Engbom and Moser (2022) by exploiting cross-state variation in the bindingness of the statutory minimum, thereby enabling me to compute marginal effects of minimum wage increases across different percentiles of the hourly wage distribution.

Taken together, my empirical findings uncover patterns consistent with the framework's predictions. Between 2012 and 2019, the real hourly minimum wage increased by around 9%. In the same period, my analysis indicates a decline in wage variance of around 6 log points and an increase in job amenity variance of around 35 log points. Although these findings do not constitute causal evidence for the wage-amenities tradeoff, I draw on the model's third prediction to strengthen the evidence of the mechanism. Consistent with the prediction, minimum wage increases between 2012 and 2019 propagate well beyond the lower tail of the wage distribution, pushing wages upward up to the 95th percentile.

In this paper, I define amenities as any non-wage feature of labor contracts, including workplace conditions, career advancement, task characteristics, or the effort required for the job.

The paper contributes to the literature on the minimum wage and its effects on labor market outcomes. A growing body of literature argues that employers can accommodate higher wages by adjusting several margins beyond employment (Clemens, 2019, 2021), with recent empirical work emphasizing two potential explanations: monopsony power (Azar et al., 2024; Berger et al., 2025; Faia et al., 2026) and passthrough to prices (Allegretto and Reich, 2018; Harasztosi and Lindner, 2019; Renkin et al., 2022). I complement this literature by showing suggestive evidence of the wage-amenities tradeoff as an additional adjustment margin, whereby firms respond to higher minimum wages by reducing job amenities.

Contrary to my approach, prior studies examine the effects of minimum wages on specific employee benefits. Clemens et al. (2018), for instance, find that workers become less likely to report employer-sponsored health insurance, while Kanayama et al. (2025) find almost no decline in flexible non-wage benefits such as transport reimbursements. The contrast in their conclusions points to a key limitation in their strategy: findings for specific benefits cannot rule out simultaneous adjustments to other non-wage attributes—especially since specific adjustments to job amenities can vary considerably across firms. My work, instead, relies on empirical predictions when employers cannot differentiate amenities across employees. This approach applies to a wide range of preferences over job amenities and compensation schemes, providing suggestive evidence that reconciles the economic intuition that firms adjust compensation in response to higher minimum wages with the recent empirical findings.[4]

Furthermore, the paper contributes to the literature on how minimum wage increases spread across the wage distribution. Previous work in the United States (Lee, 1999; Autor et al., 2016; Cengiz et al., 2019; Derenoncourt and Weil, 2025) and the United Kingdom (Stewart, 2011) indicates small, often negligible spillover effects. In contrast, Engbom and Moser (2022) demonstrate that minimum wage increases in Brazil, between 1996 and 2018, affected wages up to the 90th percentile. My framework offers an explanation for these disparate results. When employers cannot tailor job amenities to different employees, wages at the top of the distribution rise due to compensating differentials.

The remainder of the paper proceeds as follows. Section I introduces the framework and its empirical predictions on wages, job amenities, and wage floors. Section II describes the Brazilian institutional background and data sources, while Section III presents the empirical strategies used and the corresponding results.

I. Framework: Minimum wage and the wage-amenities tradeoff

The following section develops a simple framework that formalizes the tradeoff between wages and job amenities and enumerates the three main predictions I take to the data in Section III.

A. Model

The basic framework consists of a static, random, posting, partial equilibrium model in which firms choose wages and job amenities, subject to worker outside options and binding minimum wages. In this subsection, I describe the model's setting and assumptions, and characterize two distinct equilibria: one in which firms have access to flexible amenities and another in which firms allocate the same amenities to all employees. For a complete derivation of the solution, see Section A in the Appendix.

[4]Beyond research on the wage-employment tradeoff, recent studies also document negative effects on vacancies (Horton, 2025; Kudlyak et al., 2025) and separations (Dube et al., 2016), as well as worker flows toward larger, higher-paying, and more productive firms (Dustmann et al., 2021; Rao and Risch, 2026), which aligns with predictions in Board and Meyer-Ter-Vehn (2015) and Wang (2025).

Setting and assumptions. Consider a firm $j \in \mathcal{J}$ that employs two workers indexed by $i \in \{L, H\}$, where L and H denote a low-paid and a high-paid worker, respectively. Workers differ in productivity ($\alpha_L < \alpha_H$) and outside options ($u_L < u_H$), but share common preferences over wages w and amenities a, as given by $U_i = w_i^\gamma a_i$.

The firm posts worker-specific contracts $\langle w_i, a_i \rangle$. Since this is a static model, firms cannot substitute labor.[5] Therefore, if any worker rejects the contract, the firm exits and both workers receive their outside option u_i. Otherwise, the firm produces output equal to the workers' joint productivity, pays wages, and provides amenities at a unit cost p_a, earning profits equal to

$$\pi_j = f(\alpha_L, \alpha_H) - (w_L + w_H) - p_a(a_L + a_H)$$

for some generic production function $f(\alpha)$.

Total labor costs rise with wage floors. To keep the model simple, I assume that firms earn strictly positive profits, even when paying the minimum wage to all workers. An alternative interpretation is that new entrants, with a p_a low enough to break even, replace the exiting firms. Since optimal wages and job amenities do not depend on p_a for constrained firms, both interpretations are equivalent.

Under these assumptions, I can set up the firm's cost-minimization problem, subject to each worker's participation and minimum-wage constraints. Namely,

$$\begin{aligned} & \min_{\{(w_i,a_i)\}_{i=L}^{H}} && (w_L + w_H) + p_a(a_L + a_H) \\ (PC) \qquad & \text{s.t.} && w_i^\gamma a_i \geqslant u_i, \quad i \in \{L, H\} \\ (MW) \qquad & && w_i \geqslant \underline{w}, \quad i \in \{L, H\} \end{aligned}$$

Flexible amenities. When firms can differentiate job amenities among workers, each participation constraint pins down the worker's utility separately. As a result, individual optimal wages and job amenities depend on the bindingness of the corresponding minimum-wage constraint.

When the minimum wage is non-binding for worker i, the firm posts a contracts comprising the following wage and non-wage components

$$w_i^* = u_i^{\frac{1}{1+\gamma}} (p_a \gamma)^{\frac{1}{1+\gamma}} \qquad \text{and} \qquad a_i^* = u_i^{\frac{1}{1+\gamma}} (p_a \gamma)^{-\frac{\gamma}{1+\gamma}}$$

Otherwise, the firm offers the minimum wage $\underline{w}$ and reduces job amenities to alleviate the mandatory wage increase, so that $a_i^* = u_i \underline{w}^{-\gamma}$.

Higher wage floors raise labor costs for workers with a binding minimum wage, potentially leading to small employment declines that I have assumed away for simplicity. Nevertheless, note that job amenities absorb some of the upward pressure on labor costs, implying that firms will resort to layoffs less often than in the absence of job amenities.

Furthermore, the solution highlights an important aspect of the framework. As long as the firm has access to flexible amenities, variation in contracts arising from minimum wage increases and other sources—such as changes in the firm's compensation policy—are indistinguishable.

[5] This implicitly rules out firms changing their worker composition. The simplification not only focuses attention on compensation rather than workforce adjustments, but also appears empirically plausible. Workplace heterogeneity (Handwerker, 2023) among Brazilian firms remained broadly stable between 2012 and 2019. If anything, worker composition became more heterogeneous over time.

Firm-level amenities. When firms rely on a single level of amenities a for all workers, posted contracts become tightly interconnected. Even if workers earn wages above the statutory minimum, optimal contracts depend on the outside options of both low- and high-paid workers. A salient case arises when the minimum wage binds both workers. Then, the firm posts contracts paying the minimum wage and sets job amenities based on the outside option of the high-paid worker, i.e., $a^* = u_H \underline{w}^{-\gamma}$. As a result, the low-paid worker benefits from the higher wage floor and pockets a differential equal to $u_H - u_L$.

On the contrary, consider the case in which the low-paid worker is the only one with a binding wage floor. Then, the firm must offer the statutory minimum to the low-paid worker, and faces two options when setting job amenities.

On the one hand, the employer can use job amenities to satisfy the participation constraint of the high-paid worker and absorb the higher wage floor, leaving a rent for the low-paid worker. On the other hand, the firm can set jobs amenities based on the low-paid worker's outside option, i.e., $a^* = u_L \underline{w}^{-\gamma}$. By doing so, the firm negatively affects the high-paid worker, who now requires a compensating differential to remain indifferent.

Likewise to before, the employer can use job amenities to satisfy the participation constraint of the high-paid worker, absorbing part of the rise in labor costs and leaving a rent to the low-paid worker. Although in this case, the rent represents less than the gap in outside options.

On the other hand, the firm can set jobs amenities based on the low-paid worker's outside option, i.e., $a^* = u_L \underline{w}^{-\gamma}$. However, by doing so, the firm reduces the high-paid worker's total compensation, who now requires a compensating differential to remain indifferent. As a result, the firm has to post a higher wage equal to $w_H^* = \underline{w} \left(u_H / u_L\right)^{1/\gamma}$, which originates spillovers across the wage distribution. In this case, the low-paid worker does not pocket a differential, and both employees achieve utilities equal to their respective outside options.

B. Empirical predictions

One notable implication of the model is that, when firms cannot tailor job amenities to individual workers, a minimum wage increase need not compress the wage distribution nor widen the dispersion of job amenities. This contradicts the conventional wisdom that higher wage floors trim the lower tail of the wage distribution, as well as the predictions in the case of flexible amenities. However, what stands out is that without further assumptions, the model predicts effects among workers who earn above the statutory minimum.

Firms react to a minimum wage increase depending on their exposure to wage floors, which is roughly captured by the number of workers earning exactly the statutory minimum. For that reason, I define three categories. All employees at the high-paying firm earn wages above the minimum wage. Rather, low-paid workers at the medium-paying firm and all employees at the low-paying firm earn the statutory minimum.

Medium-paying firms, however, may differ in their optimal contracts depending on the unit cost of amenities and workplace heterogeneity—the gap in employees' outside options. Firms with a more homogeneous workforce (henceforth specialized employers) prefer to pass along minimum wage increases to the high-paid worker. Conversely, firms with higher workplace heterogeneity (hereafter vertical employers) favor leaving rents to low-paid employees.

Without further assumptions about the distribution of firms across categories, the model delivers empirical predictions I can map to aggregate moments observed in the data. This feature eliminates the need to track firms as they move across categories.

And, more importantly, it addresses a previous challenge in the literature: insufficient exogenous variation and limited detail on workers' contracts.

Prediction 1: Wage compression. While intuition suggests that wage floors compress the wage distribution, spillovers to higher earners make the conclusion less obvious. A significant passthrough to the average wage can compress the upper tail of the distribution while increasing dispersion at the lower tail. Therefore, a higher minimum wage can actually increase dispersion in the wage distribution by shifting its skewness upward.

The framework accounts for this possibility and concludes that, everything else equal, if specialized firms have limited workplace heterogeneity, minimum wage increases compress the wage distribution, $\downarrow \mathbb{V}(w_i)$. While not necessary in other contexts, both the assumption and the prediction align with empirical patterns in Brazilian labor markets.

Prediction 2: Increased dispersion in job amenities. While low-paying firms and specialized employers reduce job amenities to offset wage increases, the effect on the overall dispersion depends on the interplay between two opposing forces. First, the gap in job amenities compared to high-paying employers and vertical firms widens, increasing the variance. Second, the within-variance in job amenities among specialized employers and low-paying firms decreases.

The framework, however, establishes that overall dispersion increases with the minimum wage, $\uparrow \mathbb{V}(a_j)$, provided the initial dispersion in job amenities at the bottom of the distributions is limited. More importantly, in a context of continuous adjustments to the federal minimum wage—and the corresponding downward adjustments in job amenities—this scenario becomes extremely likely.

Prediction 3: Wage increases above the minimum wage. As previously shown, reducing job amenities in specialized firms negatively affects high-paid workers, who, as a result, demand that these firms pay a compensating differential to remain indifferent. As a result, an increase in the minimum wage propagates throughout the wage distribution, leading to a greater passthrough to average wages. In particular, the model anticipates spillovers up to the percentile in the wage distribution occupied by the highest-paid worker at a specialized firm.

II. Brazil: Institutional background and data

The following section describes the institutional setting and data used in the empirical analysis. Brazil provides a particularly suitable environment to study how firms adjust wages and job amenities in response to minimum wage increases. More precisely, the Brazilian labor market aligns with the context described in Section I, and the country offers high-quality matched employer-employee data that I can leverage for my analysis.

A. Labor market institutions in Brazil

Several features in the Brazilian labor market make it well-suited for my analysis. In particular, minimum wage adjustments are frequent but modest, and legal regulations can interfere with discriminating job amenities across employees. This subsection reviews relevant institutions and discusses additional empirical evidence in Brazil supporting the approach.

Minimum wage. According to the *Consolidated Labor Laws* (CLT, for its Portuguese acronym), full-time workers (44 hours per week) with a formal contract cannot earn less than the statutory minimum wage. Along the same lines, the CLT stipulates that, for part-time workers, firms must adjust the sum proportionally to the hours worked. The Brazilian government supports enforcement through mandatory annual filings and

routine labor inspections. As a result, compliance is relatively high among formal firms. And, more importantly, the minimum wage has a lighthouse effect, with a near full pass-through to informal employees working at formal firms and relatively small formal-to-informal reallocations (Saltiel and Urzúa, 2022; Derenoncourt et al., 2025).

Brazil adopted a unified federal minimum wage in 1984, although 5 out of 27 states (unidades federativas) have set independent regional minimum wage floors above the federal rate.[6] Between 2012 and 2019, minimum wage increases occurred on the first of January and followed a rule-based formula. More precisely, laws 12,382/2011 and 13,152/2015 tied annual updates of the minimum wage to past inflation and real GDP growth.

While the federal minimum wage remains the primary wage-setting institution in Brazil, unions routinely negotiate wage floors above the federal rate through collective bargaining agreements (CBA). Similar to the statutory minimum, coverage is universal. Therefore, negotiated terms apply to all employees within the same firm or industry, regardless of union membership. That said, only about 10 percent of CBA clauses concern wage-related negotiations, such as wage adjustments and floors (Lagos, 2025). Instead, a significant part of negotiations over collective agreements focuses on job amenities.

CBAs and the CLT. CBAs are binding contracts between unions and employers that apply collectively to all workers under the union's representation. Meaning that individual contracts in Brazil cannot suppress or limit rights specified in the agreement, irrespective of union membership. Moreoover, CBAs can be established at the firm level or across entire economic sectors. In sectoral CBAs, unions and employer associations negotiate general floors and conditions that build up the requirements stipulated in the CLT. A firm-level CBA, on the other hand, entails direct negotiations with the employer and addresses workplace-specific issues.[7]

The CLT is the most comprehensive labor framework in Brazil, so it often takes precedence over CBAs. It establishes minimum standards for formal employment relationships in the country and, while it does not explicitly prohibit differentiation across workers, it emphasizes uniform treatment of employees within establishments. The *Supreme Labor Court* (TST, for its Portuguese acronym) has adopted the same interpretation, ruling against firms unable to demonstrate objective, verifiable criteria for special gratifications or privileges.[8]

In comparison with labor markets in other countries, these institutional features appear to limit firms' ability to differentiate job amenities across workers, supporting the single-amenity assumption.

B. Data

The following subsection describes the data used in the empirical analysis, consisting of the *Annual Social Information Report* (RAIS, for its Portuguese acronym), and the sample selection procedures used to identify firm-level job amenities from worker flows.

[6] These five states comprise Rio de Janeiro and Rio Grande do Sul (2001), Paraná (2006), São Paulo (2007), and Santa Catarina (2010). Among them, Paraná, Rio Grande do Sul, and Santa Catarina implemented multi-layer regional wage floors that vary with occupation, industry, or qualifications. Over time, state-level minimum wages have ranged from parity with the federal rate to roughly 40 percent above it.

[7] For more information on Brazilian CBAs, see Lagos (2025). There, the author describes the nature of unions and collective agreements in Brazil, showing that, on average, up to 76% of clauses focus on job amenities unrelated to wages or separations. Furthermore, 94% of unions operate at the municipal level, negotiating sectoral and firm-level CBAs that cover around 48% and 16% of private-sector workers, respectively.

[8] For example, Summary 451/2014 ruled against a firm that excluded workers from profit-sharing bonuses according to the separation date. Similarly, Case Law Guidance 383/2011 established equal conditions for outsourced and in-house workers who performed similar tasks.

RAIS. Brazil offers high-quality data on employment relationships, establishments, and worker characteristics. Every year, the Brazilian government collects mandatory filings from registered firms, which cover the universe of formal employees. These matched employer-employee administrative records, known as RAIS, report many features central to the recent empirical literature on Brazilian labor markets (Alvarez et al., 2018; Gerard et al., 2021; Engbom and Moser, 2022; Van Doornik et al., 2023).

I focus on the period ranging from 2012 to 2019. Firms must report the exact hiring and separation dates for every worker employed in a given calendar year, allowing me to reconstruct workers' full employment histories during this period. Every employment spell details the modality, contractual weekly work hours, six-digit occupational codes, and average monthly wages, including variable pay components such as performance bonuses, profit-sharing, and overtime. More importantly, RAIS includes information about the reason notified for every hire and separation, allowing me to distinguish between layoffs, quits, and transfers. For more information about the properties of this variable, see Dobbin et al. (2025).

RAIS also contains unique individual and employer identifiers, as well as personal data (e.g., gender, age, and educational attainment) and firm characteristics (e.g., five-digit industry codes, location, and legal status). My analysis is at the establishment level, so that my definition of firm coincides with that of CBAs and the CLT. Throghout the paper, I use the terms firm, employer, and establishment interchangeably.

Sample. The analysis centers on workers aged 18 to 54 with urban open-ended contracts in the private sector. I use RAIS to construct quarterly employment histories, assigning workers with multiple contracts to their highest-earning job.

Consistent with previous empirical work, I further restrict the sample to the largest connected set of workers and firms. I estimate firm-specific wage premiums in the weakly connected set of movers (henceforth, the AKM sample) and firm-level job amenities in the strongly connected set (SCC) of actively recruited workers. A connected set defines a cluster of workers and firms linked through employment transitions over a time window. A firm belongs to the AKM sample if any of its workers were previously employed at or later moved to another firm in the weakly connected set (see Abowd et al., 1999; Card et al., 2013, 2018; Gerard et al., 2021). Similarly, a firm belongs to the SCC if it has both poached an employee from and lost an employee to another firm in the SCC through active recruiting (see Sorkin, 2018; Lehmann, 2025).

As I mentioned, estimating firm-specific wage premiums and job amenities requires defining a time window, which poses a dilemma. Longer windows increase worker mobility and improve connectedness, so the analysis covers more workers and firms, but they reduce the number of samples available to track dynamics over time. I prioritize representativeness and, therefore, split the sample into two equal periods of 16 quarters: 2012Q1-2015Q4 and 2016Q1-2019Q4.

Table 1 shows that focusing on the SCC (Columns 3 and 6) restricts the analysis to establishments that, on average, employ three to four times as many workers as the representative establishment in Brazil (Columns 1 and 4).[9] Furthermore, workers in the SCC earn higher hourly wages and show greater wage dispersion, while establishments display less dispersion in firm homogeneity.[10] Beyond these differences, other employ-

[9] Compared to existing work, four years of worker mobility in Brazil capture a larger proportion of establishments than a six-year window in the US (Sorkin, 2018) and an eight-year window in Austria (Lehmann, 2025). In Brazil, the SCC covers roughly 7 pp (percentage points) more establishments, 9 pp fewer employment relationships, and 18 pp fewer workers than the US. Compared to Austria, 12 pp more establishments, 21 pp more employment relationships, and 15 pp more workers.

[10] Firm homogeneity refers to workplace homogeneity in Handwerker (2023) and describes the overall distribution of occupations by establishment. It takes a value of 1 when employers hire only one occupation, and 0 when no two workers within the firm have the same occupation.

TABLE 1 —SUMMARY STATISTICS AND SAMPLE SELECTION

	2012-2015			2016-2019		
	RAIS	AKM	SCC	RAIS	AKM	SCC
Establishments	4,291,363	3,323,529	765,419	4,354,262	3,126,974	498,859
	100.00%	77.45%	17.84%	100.00%	71.81%	11.46%
Workers	50,416,673	49,079,671	39,338,017	46,092,427	44,137,460	30,704,508
	100.00%	97.35%	78.03%	100.00%	95.76%	66.62%
Employment spells	510,008,563	497,308,284	372,771,472	470,445,459	451,144,482	291,773,507
	100.00%	97.51%	73.09%	100.00%	95.90%	62.02%
Firm size (𝔼)	11.3	13.4	35.1	10.3	12.9	42.0
Firm homogeneity (𝔼)	0.31	0.31	0.30	0.31	0.31	0.30
Firm homogeneity (𝕍)	0.060	0.058	0.048	0.061	0.058	0.048
Log hourly wage (𝔼)	2.10	2.11	2.19	2.14	2.16	2.26
Log hourly wage (𝕍)	0.394	0.396	0.432	0.349	0.353	0.401
Hours (𝔼)	185.9	185.9	185.4	184.8	184.8	183.9
Monthly paid (%)	0.91	0.91	0.89	0.93	0.92	0.90
Female employment (%)	0.41	0.41	0.40	0.43	0.42	0.42
Age (𝔼)	33.7	33.6	33.6	34.5	34.4	34.4
Years of schooling (𝔼)	12.3	12.3	12.4	12.5	12.5	12.7

Notes: The table reports characteristics over time for employment relationships, workers, and establishments in Brazil between 2012 and 2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. The AKM sample centers on employers in the weekly connected set of movers and is used to estimate firm-specific wage premiums. The SCC focuses on employers in the strongly connected set of actively recruited employees and is used to estimate firm-level job amenities. Firm homogeneity refers to workplace homogeneity, as defined by Handwerker (2023), and displays the normalized occupational Herfindahl-Hirschman index (HHI). It measures the average occupation size within a firm, taking a value of 1 when employers hire only one occupation, and 0 when no two workers within the firm have the same occupation.

ment characteristics remain broadly similar across samples in both the 2012-2015 and 2016-2019 periods.

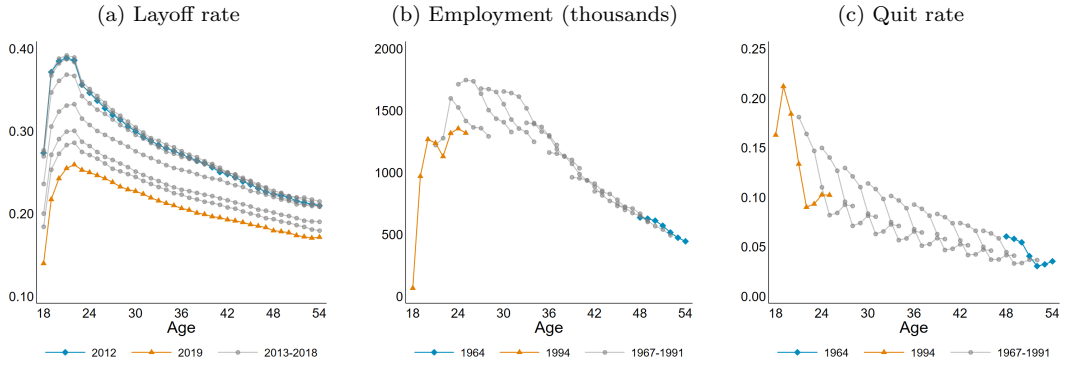
Comparing samples over time reveals that most employment characteristics exhibit only minor shifts: average monthly hours decline by roughly one hour, female participation rises by about 2 percentage points, and the average employee is one year older. By contrast, the average real hourly wage increases by 4 log points, accompanied by a 0.04 decline in its variance. These changes are stronger in the SCC, where the average rises by 7 log points and the variance declines by 0.03. Section III.A explores these distributional wage patterns in more detail.

Table 1 also shows a sharp decline of about 40 million quarterly employment relationships across the two samples. That being said, job destruction does not account for the contraction: layoff rates decline uniformly across cohorts during this period (Figure 1a). Instead, Figure 1b shows that workers born after 1991 (aged 28 or younger in 2019) exhibit lower entry rates into the formal labor market compared to older workers. At the same time, I observe a decline in worker mobility over time: quits fall uniformly across cohorts until 2016 and do not recover afterwards (Figure 1c). As a result, the SCC sample covers a smaller fraction of employment relationships, workers, and establishments in 2016-2019 compared to 2012-2015.

III. Empirical analysis: Taking the framework's predictions to the data

In this section, I turn to the empirical analysis. Results mirror the three testable implications from Section I. Namely, that higher wage floors compress the wage distribution, increase dispersion in job amenities, and raise wages above the binding minimum wage.

FIGURE 1 — EMPLOYMENT DYNAMICS IN BRAZIL (2012-2019)



Notes: The figure reports average employment dynamics in Brazil between 2012 and 2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. Panel (a) displays the annualized layoff rate by age across different years. Panel (b) shows headcounts (in thousands) of workers by age across different cohorts. Panel (c) shows the annualized quit rate by age across different cohorts. In Panel (a), each line corresponds to a year, with each dot representing the average outcome for workers of a given age that year. In Panels (b) and (c), each line conveys a cohort, with each dot representing the average outcome for workers of a given age born in that cohort. I report cohorts every three years, ranging from 1964 (48 years old in 2012) to 1994 (18 years old in 2012).

A. Reduction in wage inequality

The first prediction states that higher wage floors reduce wage inequality. Correspondingly, Brazil has experienced a long-lasting decline in wage disparities since the 1980s. As a result, numerous researchers have investigated potential explanations for this trend, although the literature has concluded that the minimum wage played a central role. Consistent with similar findings, I document a decline in wage inequality between 2012 and 2019 across a wide range of measures.

For these empirical exercises, I use the entire RAIS sample to construct five annual measures of wage dispersion. The first two variables follow standard measures in the literature. I compute the lower-tail hourly wage inequality as the ratio of the median to the 10th percentile (P50:P10), and the upper-tail hourly wage inequality as the ratio of the 90th percentile to the median (P90:P50). For the next two measures, I deviate from the standard definition by replacing the median with the average hourly wage. Lastly, I use the variance of log monthly wages.

Figure 2 uncovers that wage disparities decreased over time across all five measures. For example, Figures 2a and 2b show that the upper-tail and lower-tail hourly wage inequality fell by about 9 and 7 log points, respectively. In relation to previous studies using longer time series (Alvarez et al., 2018; Engbom and Moser, 2022), the trends during this decade indicate a more pronounced wage compression at the top of the distribution rather than at the bottom.

Figure 2d further contributes to this idea, showing that minimum hourly wage increases have had a significant passthrough to the average hourly wage. Both hourly rates grew at the same pace over 4 years; despite decoupling afterward, the average hourly wage accumulated a 4 log points growth, compared to 6 log points for the minimum hourly wage. Conversely, Figure 2c displays a sharp decline in the variance of log monthly wages as the real statutory minimum wage rose.

Firpo and Portella (2019, 2025) enumerate a series of potential explanations from the literature to account for these particular trends in Brazil. Researchers have covered many alternatives, including economic growth, productivity shocks in the tradable sector, higher educational attainment, formalization of employment, discrimination, and

unions' bargaining power, among others. Nonetheless, the consensus indicates that the minimum wage accounts for at least half of the observed decline in wage and earnings inequality.

B. Increase in job amenities inequality

The second prediction poses that higher wage floors increase the variance of job amenities. In this subsection, I first discuss the interpretation of non-wage attributes of labor contracts. Next, I present my implementation of the empirical strategy in Lehmann (2025), which I use to recover firm-specific job amenities separately for periods 2012-2015 and 2016-2019. My findings indicate a significant increase in the dispersion of job amenities in Brazil over time.

Job amenities. When workers assess the value of jobs, they weigh contracts as a combination of wage and non-wage compensation attributes. Beyond traditional employee benefits (e.g., health and life insurance, childcare, paid time off, training), there is plenty of evidence that workers value other items in their employment relationships: remote work and flexible schedules (Mas and Pallais, 2017), job stability (Wiswall and Zafar, 2018), working conditions (Maestas et al., 2023; Collis and Van Effenterre, 2025), job safety (Ferreira et al., 2025; Anelli and Koenig, 2025), or even dignity (Dube et al., 2022). For further details on the existing literature, see Mas (2025).

Along the same lines, workers may loathe certain non-wage attributes of jobs. For instance, Humlum et al. (2025) show that higher-paying firms supply worse net job amenities. Even if these employers offer better perks and flexibility to employees, these jobs also entail higher turnover, more stress, and a faster work pace. Similarly, frameworks that incorporate the wage-effort tradeoff into their models (Board and Meyer-Ter-Vehn, 2015; Wang, 2025) reach conclusions consistent with the literature on job amenities: workers are willing to pay for less demanding jobs.

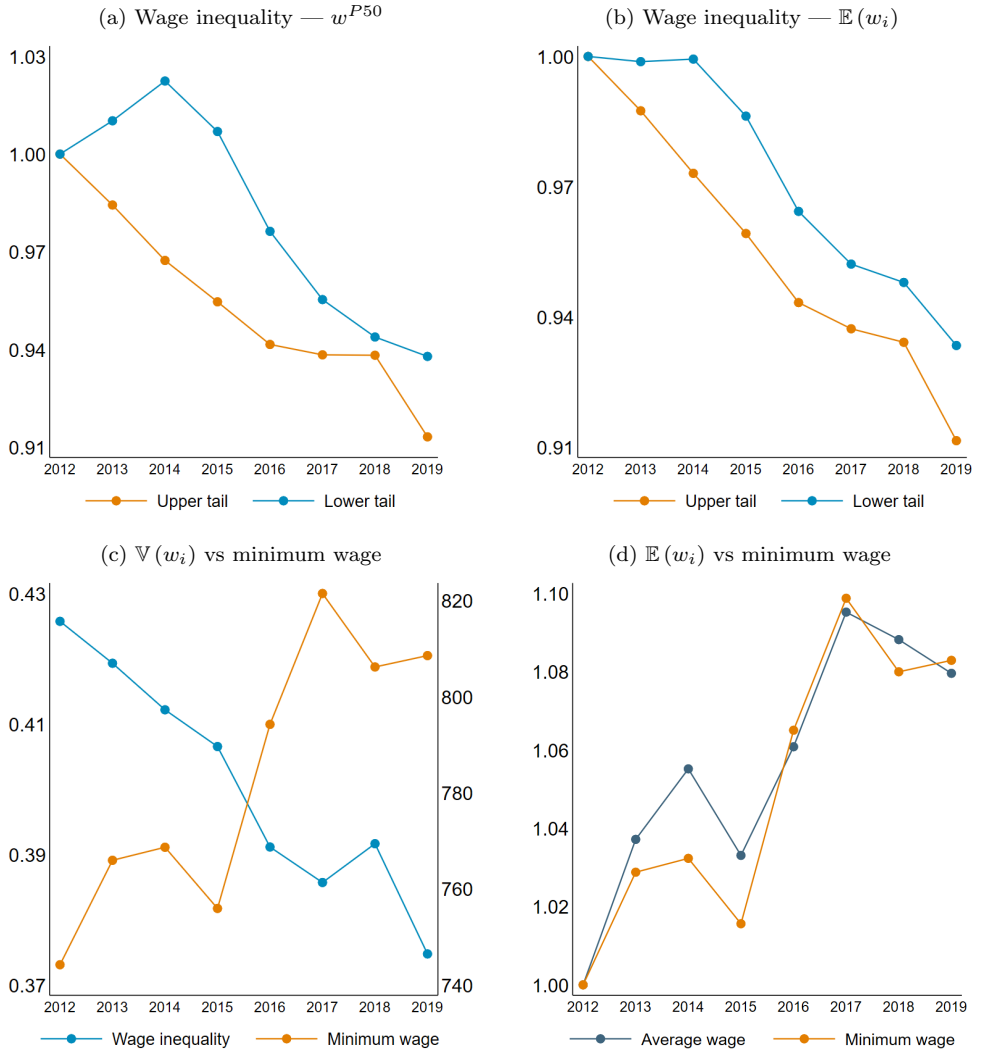
The existence of *disamenities* connects to Weil (2014), who describes a shift away from direct hiring toward alternative work arrangements. An increasing share of workers act as independent contractors, keep on-call contracts, or are employed by temporary agencies and subcontractors. Such arrangements not only preclude access to regular job amenities but can also deny benefits tied to standard employment relationships. In settings like Brazil, informality plays a similar role, putting workers in precarious conditions compared to those in formal employment.

Informal workers and independent contractors. Derenoncourt et al. (2025) demonstrate that informal workers at formal firms in Brazil almost fully benefit from minimum wage increases, despite the CLT not covering them. I argue that the wage-amenities tradeoff also applies to these individuals, but in reverse order. When firms reduce firm-level amenities to adjust for higher wage floors, they must not only compensate higher earners but also informal workers employed at the firm. Otherwise, these individuals would have incentives to leave the firm for a formal job or another better-paying informal one.[11]

I could wield a similar argument for independent contractors and workers at informal firms, even if these individuals were to keep the same working conditions in response to minimum wage increases. Nonetheless, RAIS does not cover informal workers and independent contractors in Brazil, and there is no empirical evidence at hand to support

[11] Several studies in Brazil show that formal workers eligible for unemployment insurance often agree to temporarily transition to informality (Carvalho et al., 2018; O'Leary et al., 2020; Van Doornik et al., 2023). This is a different example of the interplay between wages and the disamenities associated with informality. Here, the worker who transitions to an informal job receives a compensating differential paid by the government and not the firm: unemployment insurance.

FIGURE 2 — MINIMUM WAGE AND INEQUALITY IN BRAZIL



Notes: The figure displays summary statistics from 2012 to 2019 comparing the evolution of the minimum wage and several measures of wage inequality in Brazil. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. Panel (a) compares lower- and upper-tail hourly wage inequality over time, using the standard definition in the literature: lower-tail inequality as the ratio of the median to the 10th percentile of the outcome (P50:P10), and upper-tail inequality as the ratio of the 90th percentile to the median (P90:P50). Panel (b) repeats the empirical exercise by replacing the median with the average wage. Panel (c) compares the real minimum wage with the variance of log monthly wages over time. In the data, the correlation between the two variables is -0.906. Panel (d) compares average and minimum hourly wages over time in logs. I normalize all variables in Panels (a), (b), and (d) to one in 2012.

my argument. Thus, I cannot formally extend my analysis to these workers, although I do not expect this to affect the overall interpretation of the results.

Estimation of firm-level amenities. The backbone of the empirical strategy is Sorkin (2018). As in the framework in Section I, it assumes a partial equilibrium model with posting. These assumptions imply that firm heterogeneity is taken as given and that there is no bargaining. In addition, the model introduces further assumptions about how workers search for jobs. First, employed and non-employed workers receive the same number of offers from an identical distribution—meaning, a non-directed, on-the-job search with exogenous effort. Second, workers value firms the same, up to i.i.d. idiosyncratic preferences, which translates into a unique firm ranking—or a uniform job ladder across all workers.

Employers offer contracts $\langle w_{it}, a_j \rangle$ and differ according to size g_j, exogenous separation rates δ_j and ρ_j, as well as exogenous hiring rates f_j.

A worker i values a job at firm j from the flow payoff $v_j(w_{it}, a_j)$ and the corresponding continuation value, which they discount at rate β. With probability δ_j, the firm lays off the worker. Then, with probability ρ_j, the worker finds a job before the end of the next quarter. Otherwise, the layoff leads to nonemployment. If there is no exogenous job destruction, the worker receives an offer from firm k with probability λ_E and decides whether to quit and accept it or to stay at their current firm. With probability $1-\lambda_E$, the worker does not receive any new offer but can still quit and move to nonemployment if preferred.

$$
\begin{aligned}
\underbrace{V_j^e(w_{it}, a_{it})}_{\text{value of being at j}} &= \underbrace{v_j(w_{it}, a_{it})}_{\substack{\text{flow}\\ \text{payoff}}} + \beta \cdot \mathbb{E}\Bigg[\underbrace{\delta_j(1-\rho_j) \int_{\iota_1} \{V^n + \iota_1\}\, dI}_{\substack{\text{exogenous}\\ \text{job destruction}}} + \underbrace{\delta_j \rho_j \sum_k \int_{\iota_2} \{V_k^e + \iota_2\}\, dI\, \tilde{f}_k}_{\text{exogenous employer-to-employer}} \\
&+ \underbrace{(1-\delta_j)}_{\substack{\text{no exogenous}\\ \text{job destruction}}} \times \Bigg(\underbrace{\overbrace{\lambda_E}^{\text{offer}} \sum_k \int_{\iota_3} \int_{\iota_4} \max\{\overbrace{V_k^e + \iota_3}^{\text{accept}}, \overbrace{V_j^e + \iota_4}^{\text{reject}}\}\, dI\, dI f_k}_{\text{endogenous employer to employer}} \\
&+ \underbrace{\overbrace{(1-\lambda_E)}^{\text{no offer}} \int_{\iota_5} \int_{\iota_6} \max\{\overbrace{V^n + \iota_5}^{\text{leave}}, \overbrace{V_j^e + \iota_6}^{\text{stay}}\}\, dI\, dI f_k}_{\text{endogenous job destruction}} \Bigg) \Bigg]
\end{aligned}
$$

The decision to quit a firm to take a job at a new one is what reveals workers' preferences for employers. As Sorkin (2018) shows, the relative flows across any pair of firms determine the relative value between the two employers. Moreover, one does not need to observe flows across every pair of firms; it is sufficient for the firm to be part of the SCC.

$$
\underbrace{\frac{M_{jk}}{M_{kj}}}_{\substack{\text{relative}\\ \text{flows}}} = \underbrace{\frac{f_j}{f_k}}_{\substack{\text{relative}\\ \text{offers}}} \underbrace{\frac{g_k(1-\delta_k)}{g_j(1-\delta_j)}}_{\substack{\text{effective relative}\\ \text{firm size}}} \underbrace{\frac{\mathbb{P}(j \succ k)}{\mathbb{P}(k \succ j)}}_{\substack{\text{relative}\\ \text{firm values}}}
$$

From this point forward, I follow Lehmann (2025) and assume that workers employed at firm j derive utility from wages and job amenities as

$$
V_{it}^j = V_j^e(w_{it}, a_{it}) = w_{it}^\gamma \cdot a_{it} \tag{1}
$$

where γ represents the elasticity of substitution between wages and job amenities. Furthermore, I assume that a_{it} depends on firm-specific job amenities $\chi_{J(it)}$ and id-

iosyncratic preferences ι_{it},

$$\log \ a_{it} = \chi_{J(it)} + \iota_{it}, \quad \iota \sim N(0, \sigma_\iota^2) \tag{2}$$

while wages are jointly determined by worker and firm characteristics, following an AKM model specification (Abowd et al., 1999; Card et al., 2013, 2018)

$$\log \ w_{it} = \alpha_i + \psi_{J(it)} + X_{it} \cdot b + \nu_{it}, \quad \nu \sim N(0, \sigma_\nu^2) \tag{3}$$

I estimate this model of wage determination separately for 2012-2015 and 2016-2019 using the AKM sample. In Section B in the Appendix, I provide more details on the procedure and compare estimations across samples and over time.

By replacing Equations (2) and (3) into (1), I can express the worker's utility function in terms of fixed worker and firm characteristics, as well as idiosyncratic i.i.d. shocks to wages and preferences for job amenities, $\varepsilon_{it} = \gamma\nu_{it} + \iota_{it}$.

$$\log \ V_{it}^j = \gamma\left(\alpha_i + \psi_j + X_{it}\beta\right) + \chi_j + \underbrace{\gamma\nu_{it} + \iota_{it}}_{\varepsilon}, \quad \varepsilon_{it} \sim N(0, \underbrace{\gamma^2\sigma_\nu^2 + \sigma_\iota^2}_{\sigma_\varepsilon^2})$$

A worker i who receives an offer from firm k must make a maximizing decision by comparing V_{it}^j and V_{it}^k, which is why economists consider these transitions as *endogenous*.[12] When a worker makes a voluntary move, it reveals that she ranks compensation in firm k higher than in firm j on the job ladder.

$$\begin{aligned}
\mathbb{P}(k \succ j) &= \mathbb{P}\left(\log V_{it}^k > \log V_{it}^j\right) \\
&= \mathbb{P}\Big(\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j) > (\varepsilon_{1it} - \varepsilon_{2it})\Big) \\
&= \Phi\left(\frac{\gamma\big(\psi_k - \psi_j\big) + \big(\chi_k - \chi_j\big)}{2\sigma_\varepsilon}\right)
\end{aligned}$$

Identification relies on the wage-amenities tradeoff discussed in Section I. In essence, if a worker values compensation at two firms equally, $V_{it}^j = V_{it}^k$, any difference in firm-specific pay premiums, $\psi_j - \psi_k$, can be attributed to compensating differentials due to uneven job amenities, $\chi_j - \chi_k$, or idiosyncratic taste differences, $\varepsilon_{1it} - \varepsilon_{2it}$.

In equilibrium, better-skilled workers, on average, earn higher wages and benefit from more generous job amenities at the firm. However, the contracts of individuals with similar skills may vary greatly across employers. Intuitively, all these contracts lie over the same indifference curve, as workers ultimately derive the same utility from any of them. Note that this is based on a partial equilibrium model; as such, I abstract from why contracts differ across employers and treat these choices as given.

To take this relationship to the data, I specify the following likelihood function

$$\mathcal{L} = \prod_{n=1}^{N} \Phi\left(\frac{\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j)}{2\sigma_\varepsilon}\right)^{\frac{1}{g_j}\frac{1}{f_k}}$$

where g_j and f_k account for an important heterogeneity across firms. Consider the case where firms j and k are identical except that k makes twice as many job offers as j.

[12] A benefit of RAIS is that I can directly tell quits apart from layoffs, solving the problem of distinguishing endogenous from exogenous job-to-job transitions. For instance, Lehmann (2025) relies on unemployment benefit records, and Sorkin (2018) targets excess layoffs during downturns to calibrate the probability that any job-to-transition was a quit.

Then, one should expect twice as many workers moving towards firm k as to firm j. Likewise, if k employed twice as many workers as j, one should expect twice as many workers leaving firm k compared to firm j.

The estimation does not yield estimates for γ and $\chi_j,\ \forall j \in \mathcal{J}$. Notice that without assumptions on the scaling parameter σ_ϵ, the identification of these parameters becomes unfeasible. Instead, I obtain the following estimands

$$\hat{\gamma} = \frac{\gamma}{2\sigma_\varepsilon} \quad \text{and} \quad \hat{\chi}_j = \frac{\chi_j}{2\sigma_\varepsilon}$$

However, the coefficient given by the ratio between the two

$$\hat{\chi}^{BRL} = \hat{\chi}/\hat{\gamma} = \chi/\gamma$$

does not depend on σ_ϵ and, as such, is exempt from this limitation. In addition, the expression offers a nice interpretation, as $\hat{\chi}^{BRL}$ represents the value of firm-specific job amenities in log-BRL (*reais brasileiros*). For more details on the empirical estimation of firm-specific job amenities and how to account for noisy estimates in the plug-in estimator of the variance, see Section C in the Appendix.

Figure 3 plots the variance of job amenities in Brazil, adjusted for firm size, across 2012-2015 and 2016-2019, showing a significant increase in the dispersion of firm-specific job amenities over time. Point estimates rose from 0.0628 in 2012-2015 to 0.4052 in 2016-2019. I bootstrap the likelihood function 50 times to compute confidence intervals for the variance estimations. The difference remains statistically significant at the 95% level, even when accounting for the later period being less precisely estimated. Although this result does not imply causality, the empirical exercise supports the framework's intuition that firms reduce job amenities in response to wage floor increases, thereby increasing job amenities inequality nationwide.

C. Spillovers across the entire wage distribution

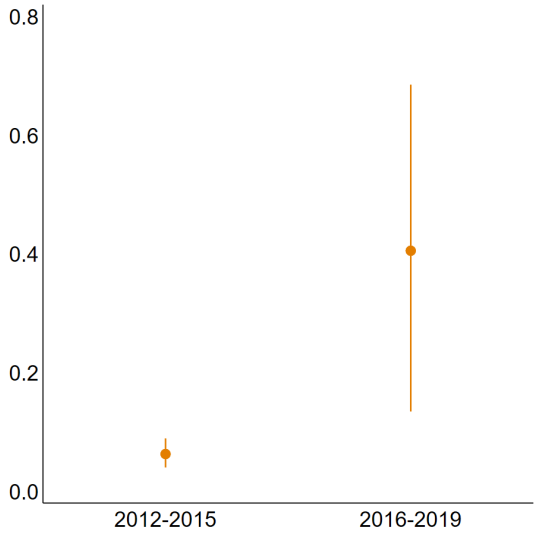
The third prediction indicates that medium-paying firms generate wage spillovers throughout the wage distribution. The empirical strategy in this subsection leverages cross-state variation in the bindingness of the federal minimum wage, namely the *effective minimum wage*. I follow Engbom and Moser (2022) to estimate the marginal effects of minimum wage increases across different percentiles of the wage distribution. In particular, I perform the analysis separately for the periods 2012-2015 and 2016-2019, finding that higher wage floors propagate up to the 85th and 95th percentile of the wage distribution, respectively.

Consider a firm that pays the minimum wage to a fraction of its workforce, while the remaining employees earn above that level. When the firm reduces job amenities to accommodate a higher wage floor, workers who earned above the new statutory minimum now perceive a loss in their total compensation. In consequence, the firm has to offer compensating differentials, and wage increases propagate across the wage distribution.

Lee (1999) develops an empirical strategy to measure wage spillovers separately from changes in the latent wage distribution. That is, the underlying wage distribution that one would observe in the absence of wage floors. This is important; otherwise, one could attribute wage changes to the minimum wage when they are actually due to other concurrent events.

The strategy defines measures for the effective minimum wage and wage dispersion relative to an arbitrary point in the wage distribution, p', and exploits cross-state

FIGURE 3 — VARIANCE OF JOB AMENITIES IN BRAZIL



Notes: The figure compares the variance of firm-level job amenities in Brazil over time. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. I estimate the variance of firm-level job amenities for establishments in the SCC, separately for the periods 2012-2015 and 2016-2019. Since estimates of job amenities contain noise, I obtain the variance of the estimation error by bootstrapping the likelihood function 50 times and use it to adjust the variance of firm-level job amenities using a formula (Sorkin, 2018; Lehmann, 2025). To obtain confidence intervals, I correct variance estimates across all 50 bootstraps and report the 5th and 95th percentiles of the distribution. See Section C in the Appendix for details.

variation in the former. As long as p' remains completely unaffected by the minimum wage, co-movements between the two variables imply direct effects or spillovers related to the minimum wage.

The first step is to establish the effective minimum wage in Brazil. I define the variable $kaitz_{st}(p') \equiv \log\ w_t^{min} - \log\ w_{st}(p')$, which stands for the log difference between the federal minimum hourly wage w_t^{min} and the pth percentile of the hourly wage distribution in state s and quarter t. Thus, as the bindingness becomes tighter, $kaitz_{st}(p')$ becomes less negative.

The next step is to generate different measures of wage dispersion across wage percentiles $p \in \{5, 10, ..., 95\}$. Thus, I define $y_{st}(p; p') \equiv \log\ w_{st}(p) - \log\ w_{st}(p')$, which represents the log difference between wage percentiles p and p' in state s and quarter t. Intuitively, as the wage distribution compresses, $y_{st}(p; p')$ moves towards zero.

I repeat this process separately for the periods 2012-2015 and 2016-2019, which allows me to construct two balanced panels at the state-quarter level, each comprising 27 states and 16 quarters.

Then, I use the following specification

$$y_{st}(p; p') = \sum_{n=1}^{2} \beta_n(p) \times kaitz_{st}(p')^n + \delta_s(p) + \sum_{n=1}^{2} b_s^n(p) \times t^n$$

to estimate non-linear effects of minimum wage's bindingness on wage dispersion, separately for each wage percentile p. I include linear and quadratic measures of effective minimum wage, as well as state fixed effects and state-specific linear and quadratic time trends. Since the specification is quadratic, I compute the marginal effects $\tau(p)$

for each wage percentile $p \in \{5, 10, ..., 95\}$ using

$$\tau(p) = \beta_1(p) + 2 \cdot \beta_2(p)\, \overline{kaitz_{st}(p')}$$

where $\overline{kaitz_{st}(p')}$ is the average $kaitz_{st}(p')$ across states and quarters.

Before moving forward, there are two things worth mentioning. First, previous studies using the same empirical approach in the US set $p' = 50$, arguing that the median wage is unlikely to be directly or indirectly affected by the minimum wage (Lee, 1999; Autor et al., 2016). On the contrary, Engbom and Moser (2022) show that spillovers in Brazil are significant and opt for $p' = 90$. In this paper, I lean toward $p' = 95$.

Second, Autor et al. (2016) claim the specification is weak to measurement error and potential transitory shocks correlated to $kaitz_{st}(p')$. Therefore, they propose an IV strategy that uses three instruments: the log real minimum wage, its square, and its interaction with the state-specific average over time of the percentile p' of the log real hourly wage distribution, $\overline{w(p')}_s$. According to the authors, the minimum wage level should affect wage dispersion only through the concurrent minimum wage's bindingness at the state level, which satisfies the exclusion restriction. I implement their IV strategy by replacing the log real minimum wage with its hourly counterparts.

Figure 4 reports the estimated marginal effects $\tau(p)$ for $p \in \{5, 10, ..., 95\}$, showing significant spillovers up to the top of the wage distribution. For example, during 2012-2015, my findings indicate positive spillovers up to the 80th percentile of the hourly wage distribution. Between 2016 and 2019, spillovers appear to be more significant than in the preceding years. Notice that the blue line (2016-2019) lies above the orange line (2012-2015) for all percentiles from the median to the 90th percentile. A plausible explanation is that compensating differentials became more pronounced at the top of the distribution due to larger reductions in job amenities in the last four years.

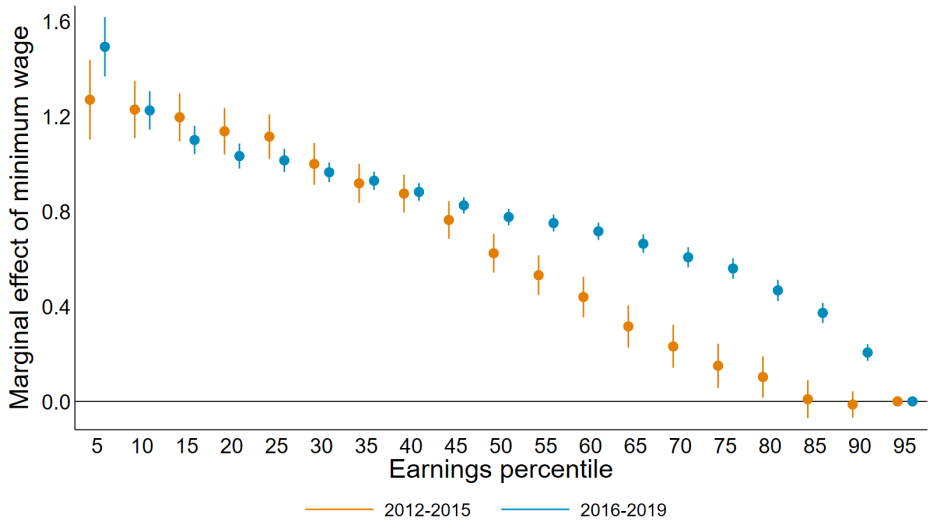
IV. Discussion

This paper supports the thesis that the minimum wage in Brazil has triggered adjustments beyond wages at the bottom of the distribution. First, the higher dispersion in job amenities between 2016 and 2019 is consistent with firms reducing non-wage compensation in response to higher wage floors. And, second, the significant spillovers at the very top of the hourly wage distribution suggest that firms paid compensating differentials to offset the decline in job amenities.

These results contribute to the literature on the minimum wage. On the one hand, it spotlights the role of *job amenities* as an important margin that firms can resort to instead of employment terminations. This channel joins other explanations in the literature—monopsony power and passthrough to prices—capable of bridging the economic intuition that firms adjust to higher wage floors with the empirical findings that show no employment declines. On the other hand, the chapter offers a reasonable explanation of why minimum wage increases in Brazil propagate through the entire wage distribution, but similar patterns do not emerge in other economies. Since Brazilian employers cannot tailor job amenities to different employees, wages at the top of the distribution rise due to compensating differentials within the firm.

Nevertheless, it remains unclear which job amenities firms may have cut back on over the decade. Notwithstanding, available evidence points to certain job amenities staying unscathed. First, reductions probably did not extend to non-wage attributes covered in CBAs. As Lagos (2025) explains, unions accumulated substantial bargaining power in 2012, following a TST ruling that automatically extended benefits from overdue CBAs. Second, some perks appear to be costless or byproducts of broader organizational decisions, making them unlikely for firms to cut deliberately. For example, a 2014 survey

FIGURE 4 — ESTIMATED MINIMUM WAGE EFFECTS ON THE WAGE DISTRIBUTION



Notes: The figure plots the estimated marginal effects of minimum wage increases across different wage distribution percentiles in Brazil. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. I follow Lee (1999) and Engbom and Moser (2022) by exploiting cross-state variation in wage floor bindingness. The baseline specification includes state fixed effects and state-specific linear and quadratic quarterly time trends. I replicate the instrumental variables (IV) strategy in Autor et al. (2016) by predicting wage floor bindingness using the log minimum real hourly wage, its square, and its interaction with the state-specific average over time of the percentile p' of the log real hourly wage distribution. I estimate marginal effects using the entire RAIS sample, separately for the periods 2012-2015 and 2016-2019. Each dot represents a marginal effect and corresponds to a separate regression of the dependent variable, ranging from the 5th to the 95th percentile of the wage distribution. Bars represent 95 percent confidence intervals based on regular standard errors.

in Brazil indicates that workers place great value on professional relationships at work, recognition, and flexibility (The Boston Consulting Group, 2016).

The same survey, however, enumerates other job characteristics that workers value and can be costly to provide, e.g., training, work-life balance, career progression, or paid timeoff. While all these job amenities provide firms with a natural margin of adjustment to offset higher wage floors, work-life balance covers broader aspects worth discussing. On the one hand, firms may be making jobs more demanding. Board and Meyer-Ter-Vehn (2015) anticipate in their model that the minimum wage acts as a coordination mechanism, creating a mass of jobs that require more effort in exchange for the higher wage floor. Although the mechanism is absent in my framework—by assuming fixed worker productivity—, my empirical approach captures disamenities related to overwork, such as stress, multitasking, and extra hours.

Along the same lines, jobs may stay equally challenging but shift towards routine and boredom. For instance, 40% of all independent contractors and 29% of all employees surveyed in Brazil report dissatisfaction with their current work situation, with higher figures among short-tenured workers and those reporting low motivation. Moreover, 67% of demotivated workers are actively seeking new positions, indicating their intention to pursue a more suitable job elsewhere (The Boston Consulting Group, 2016).

References

Abowd, John M., Francis Kramarz, and David N. Margolis. 1999. "High wage workers and high wage firms." *Econometrica*, 67(2): 251–333.

Abowd, John M., Kevin L. McKinney, and Ian M. Schmutte. 2019. "Modeling endogenous mobility in earnings determination." *Journal of Business & Economic Statistics*, 37(3): 405–418.

Addison, John T., McKinley L. Blackburn, and Chad D. Cotti. 2009. "Do minimum wages raise employment? Evidence from the U.S. retail-trade sector." *Labour Economics*, 16(4): 397–408.

Addison, John T., McKinley L. Blackburn, and Chad D. Cotti. 2012. "The effect of minimum wages on labour market outcomes: county-level estimates from the restaurant-and-bar sector." *British Journal of Industrial Relations*, 50(3): 412–435.

Allegretto, Sylvia, and Michael Reich. 2018. "Are local minimum wages absorbed by price increases? Estimates from internet-based restaurant menus." *Industrial and Labor Relations Review*, 71(1): 35–63.

Allegretto, Sylvia, Arindrajit Dube, Michael Reich, and Ben Zipperer. 2017. "Credible research designs for minimum wage studies: A response to Neumark, Salas, and Wascher." *Industrial and Labor Relations Review*, 70(3): 559–592.

Alvarez, Jorge, Felipe Benguria, Niklas Engbom, and Christian Moser. 2018. "Firms and the decline in earnings inequality in Brazil." *American Economic Journal: Macroeconomics*, 10(1): 149–189.

Anelli, Massimo, and Felix Koenig. 2025. "Willingness to pay for workplace amenities."

Autor, David H., Alan Manning, and Christopher L. Smith. 2016. "The contribution of the minimum wage to US wage inequality over three decades: A reassessment." *American Economic Journal: Applied Economics*, 8(1): 58–99.

Azar, José, Emiliano Huet-Vaughn, Ioana Marinescu, Bledi Taska, and Till Von Wachter. 2024. "Minimum wage employment effects and labour market concentration." *Review of Economic Studies*, 91(4): 1843–1883.

Berger, David, Kyle Herkenhoff, and Simon Mongey. 2025. "Minimum wages, efficiency, and welfare." *Econometrica*, 93(1): 265–301.

Board, Simon, and Moritz Meyer-Ter-Vehn. 2015. "Relational contracts in competitive labour markets." *The Review of Economic Studies*, 82(2): 490–534.

Bonhomme, Stéphane, Thibaut Lamadon, and Elena Manresa. 2019. "A distributional framework for matched employer employee data." *Econometrica*, 87(3): 699–739.

Brown, Charles, Curtis Gilroy, and Andrew Kohen. 1982. "The effect of the minimum wage on employment and unemployment: A survey."

Card, David, Ana Rute Cardoso, and Patrick Kline. 2016. "Bargaining, sorting, and the gender wage gap: Quantifying the impact of firms on the relative pay of women." *The Quarterly Journal of Economics*, 131(2): 633–686.

Card, David, Ana Rute Cardoso, Joerg Heining, and Patrick Kline. 2018. "Firms and labor market inequality: Evidence and some theory." *Journal of Labor Economics*, 36(S1): S13–S70.

Card, David, and Alan B. Krueger. 1994. "Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania." *American Economic Review*, 84(4): 772–793.

Card, David, and Alan B. Krueger. 1995. "Time-series minimum-wage studies: A meta-analysis." *American Economic Review*, 85(2): 238–243.

Card, David, and Alan B. Krueger. 2000. "Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania: Reply." *American Economic Review*, 90(5): 1397–1420.

Card, David, and Alan B. Krueger. 2016. *Myth and measurement: The new Economics of the minimum wage.* New Jersey, US:Princeton University Press.

Card, David, Jörg Heining, and Patrick Kline. 2013. "Workplace heterogeneity and the rise of West German wage inequality." *The Quarterly Journal of Economics*, 128(3): 967–1015.

Carvalho, Cristiano C., Raphael Corbi, and Renata Narita. 2018. "Unintended consequences of unemployment insurance: Evidence from stricter eligibility criteria in Brazil." *Economics Letters*, 162: 157–161.

Cattaneo, Matias D., Richard K. Crump, Max H. Farrell, and Yingjie Feng. 2025. "Binscatter regressions." *The Stata Journal*, 25(1): 3–50.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The effect of minimum wages on low-wage jobs." *The Quarterly Journal of Economics*, 134(3): 1405–1454.

Clemens, Jeffrey. 2019. "Making sense of the minimum wage: A roadmap for navigating recent sesearch."

Clemens, Jeffrey. 2021. "How do firms respond to minimum wage increases? Understanding the relevance of non-employment margins." *Journal of Economic Perspectives*, 35(1): 51–72.

Clemens, Jeffrey, and Michael Strain. 2021. "The heterogeneous effects of large and small minimum wage changes: Evidence over the short and medium run using a pre-analysis plan."

Clemens, Jeffrey, Lisa Kahn, and Jonathan Meer. 2018. "The minimum wage, fringe benefits, and worker welfare."

Collis, Manuela R., and Clémentine Van Effenterre. 2025. "Workplace hostility."

Derenoncourt, Ellora, and David Weil. 2025. "Voluntary minimum wages: The local labor market effects of national retailer policies." *The Quarterly Journal of Economics*, 140(3): 1901–1958.

Derenoncourt, Ellora, François Gerard, Lorenzo Lagos, and Claire Montialoux. 2025. "Minimum wages and informality."

Dobbin, Cauê, Daniel Fernandez, and Tom Zohar. 2025. "Separations revisited: Do layoffs or quits drive lower separation rates in high-quality firms?"

Dube, Arindrajit, Suresh Naidu, and Adam Reich. 2022. "Power and dignity in the low-wage labor market: Theory and evidence from Wal-Mart workers."

Dube, Arindrajit, T. William Lester, and Michael Reich. 2010. "Minimum wage effects across State borders: Estimates using contiguous counties." *Review of Economics and Statistics*, 92(4): 945–964.

Dube, Arindrajit, T. William Lester, and Michael Reich. 2016. "Minimum wage shocks, employment flows, and labor market frictions." *Journal of Labor Economics*, 34(3): 663–704.

Dustmann, Christian, Attila Lindner, Uta Schönberg, Matthias Umkehrer, and Philipp Vom Berge. 2021. "Reallocation effects of the minimum wage." *The Quarterly Journal of Economics*, 137(1): 267–328.

Engbom, Niklas, and Christian Moser. 2022. "Earnings inequality and the minimum wage: Evidence from Brazil." *American Economic Review*, 112(12): 3803–3847.

Faia, Ester, Benjamin Lochner, and Benjamin Schoefer. 2026. "Monopsony, markdowns, and minimum wages."

Ferreira, Susana, Sara Martínez De Morentin, and Amaya Erro-Garcés. 2025. "Measuring job risks when hedonic wage models do not do the job." *Journal of Environmental Economics and Management*, 130(103120): 1–21.

Firpo, Sergio, and Alysson Portella. 2019. "Decline in wage inequality in Brazil : A survey."

Firpo, Sergio, and Alysson Portella. 2025. "Declining wage inequality in developing countries: The case of Brazil." *Journal of Economic Literature*, 63(3): 875–915.

Gerard, François, Lorenzo Lagos, Edson Severnini, and David Card. 2021. "Assortative matching or exclusionary hiring? The impact of employment and pay policies on racial wage differences in Brazil." *American Economic Review*, 111(10): 3418–3457.

Hahn, Jinyong, and Whitney Newey. 2004. "Jackknife and analytical bias reduction for nonlinear panel models." *Econometrica*, 72(4): 1295–1319.

Handwerker, Elizabeth Weber. 2023. "Outsourcing, occupationally homogeneous employers, and wage inequality in the United States." *Journal of Labor Economics*, 41(S1): S173–S203.

Harasztosi, Peter, and Attila Lindner. 2019. "Who pays for the minimum wage?" *American Economic Review*, 109(8): 2693–2727.

Horton, John J. 2025. "Price floors and employer preferences: Evidence from a minimum wage experiment." *American Economic Review*, 115(1): 117–146.

Humlum, Anders, Mette Rasmussen, and Evan Rose. 2025. "Firm premia and match effects in pay vs. amenities."

Hurst, Erik, Patrick Kehoe, Elena Pastorino, and Thomas Winberry. 2022. "The distributional impact of the minimum wage in the short and long run."

Kanayama, Hayato, Sho Miyaji, and Suguru Otani. 2025. "Who bears the cost? High-frequency evidence on minimum wage effects and amenity pass-through in spot labor markets."

Kline, Patrick, Raffaele Saggio, and Mikkel Sølvsten. 2020. "Leave-out estimation of variance components." *Econometrica*, 88(5): 1859–1898.

Kudlyak, Marianna, Murat Tasci, and Didem Tüzemen. 2025. "Minimum wage increases and vacancies." *Labour Economics*, 97(102765): 1–12.

Lagos, Lorenzo. 2025. "Union bargaining power and the amenity-wage tradeoff."

Lee, D. S. 1999. "Wage inequality in the United States during the 1980s: Rising dispersion or falling minimum wage?" *The Quarterly Journal of Economics*, 114(3): 977–1023.

Lehmann, Tobias. 2025. "Non-wage job values and implications for inequality."

Lentz, Rasmus, Suphanit Piyapromdee, and Jean-Marc Robin. 2023. "The anatomy of sorting—Evidence from Danish data." *Econometrica*, 91(6): 2409–2455.

Maestas, Nicole, Kathleen J. Mullen, David Powell, Till Von Wachter, and Jeffrey B. Wenger. 2023. "The value of working conditions in the United States and implications for the structure of wages." *American Economic Review*, 113(7): 2007–2047.

Mas, Alexandre. 2025. "Non-wage amenities."

Mas, Alexandre, and Amanda Pallais. 2017. "Alternative work arrangements." *American Economic Review*, 107(12): 3722–3759.

Neumark, David. 2017. "The employment effects of minimum wages: Some questions we need to answer."

Neumark, David. 2019. "The econometrics and economics of the employment effects of minimum wages: Getting from known unknowns to known knowns." *German Economic Review*, 20(3): 293–329.

Neumark, David, and William L. Wascher. 2008. *Minimum Wages.*

Neumark, David, and William Wascher. 2000. "Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania: Comment." *American Economic Review*, 90(5): 1362–1396.

Neumark, David, J. M. Ian Salas, and William Wascher. 2014. "Revisiting the minimum wage—employment debate: Throwing out the baby with the bathwater?" *Industrial and Labor Relations Review*, 67(3_suppl): 608–648.

O'Leary, Christopher J., Túlio Cravo, Ana Cristina Sierra, and Leandro Justino Veloso. 2020. "Effects of unemployment insurance reforms in Brazil."

Rao, Nirupama L, and Max Risch. 2026. "Who's afraid of the minimum wage? Measuring the impacts on independent businesses using matched US tax returns." *The Quarterly Journal of Economics*, 141(1): 373–427.

Renkin, Tobias, Claire Montialoux, and Michael Siegenthaler. 2022. "The pass-through of minimum eages into US retail prices: Evidence from supermarket scanner data." *The Review of Economics and Statistics*, 104(5): 890–908.

Saltiel, Fernando, and Sergio Urzúa. 2022. "Does an increasing minimum wage reduce formal sector employment? Evidence from Brazil." *Economic Development and Cultural Change*, 70(4): 1403–1437.

Sockin, Jason. 2022. "Show me the amenity: Are higher-paying firms better all around?"

Song, Jae, David J Price, Fatih Guvenen, Nicholas Bloom, and Till Von Wachter. 2019. "Firming up inequality." *The Quarterly Journal of Economics*, 134(1): 1–50.

Sorkin, Isaac. 2018. "Ranking firms using revealed preference." *The Quarterly Journal of Economics*, 133(3): 1331–1393.

Stewart, Mark B. 2011. "Wage inequality, minimum wage effects, and spillovers."

Stigler, George J. 1946. "The economics of minimum wage legislation." *American Economic Review*, 36(3): 358–365.

The Boston Consulting Group. 2016. "Understanding Brazil's workforce in a troubled time." Sao Paulo, Brasil.

Van Doornik, Bernardus, David Schoenherr, and Janis Skrastins. 2023. "Strategic formal layoffs: Unemployment insurance and informal labor markets." *American Economic Journal: Applied Economics*, 15(1): 292–318.

Wang, Tong. 2025. "Labour market outcomes of minimum wage changes: A behavioural perspective." *Labour Economics*, 97(102824): 1–15.

Weil, David. 2014. *The fissured workplace: Why work became so bad for so many and what can be done to improve it.* Massachusetts, US:Harvard University Press.

Wiswall, Matthew, and Basit Zafar. 2018. "Preference for the workplace, investment in human capital, and gender." *The Quarterly Journal of Economics*, 133(1): 457–507.

Wolfson, Paul, and Dale Belman. 2019. "15 Years of research on US employment and the minimum wage." *Labour*, 33(4): 488–506.

A. The framework

The framework consists of a static, random, posting, partial equilibrium model. Random means that matches happen by chance. Posting means that there is no bargaining. Partial equilibrium and posting mean that I take firm and worker heterogeneity as given. Static and random mean that firms cannot substitute labor but can exit to avoid losses. Firms design contracts including wages and job amenities, subject to binding minimum wages. Workers can accept or reject the offer. The following section describes the model's setting and assumptions, characterizes separate solutions depending on whether firms can tailor job amenities to each worker, and develops empirical predictions in the case that firms set firm-level job amenities.

1. The model

Setting. Consider an economy with J firms and $2J$ workers. Each firm employs a team of two workers, with individuals being heterogeneous in both productivity α and outside option u. Firms differ in productivity solely because of their internal organization, implying that better-run firms are matched with more productive teams.

For convenience, I index agents so that firm $j \in \mathcal{J}$ matches with workers j and $2j$, and relabel them so that $j = L(j)$ and $2j = H(j)$, with $\{L(j), H(j)\} \subset \mathcal{I}$. I impose that $\mathcal{I}$ is a partially ordered set, with $\alpha_{H(j)} > \alpha_{L(j)} > 0$ and $u_{H(j)} > u_{L(j)} > 0$, but without an established relationship across firms. Therefore, $L(j)$ and $H(j)$ designate low- and high-productivity workers, respectively, within the firm only.

At the start of the period, each firm j matches with its corresponding team, with workers indexed by $i \in \{L(j), H(j)\}$. Immediately thereafter, the firm posts worker-specific contracts $\langle w_i, a_i \rangle$, consisting of wages w_i and amenities a_i. Workers share preferences over these attributes, attaining a utility equal to

$$U_i = w_i^{\gamma} a_i$$

where $\gamma > 0$ symbolizes the intensity of preference for wages.

If both workers accept the offer, the firm generates revenues equal to $f\left(\alpha_{L(j)}, \alpha_{H(j)}\right)$, pays wages, and provides amenities at a firm-specific unitary cost $p_a(j)$. So, the firm earns profits equal to

$$\pi_j = f\left(\alpha_{L(j)}, \alpha_{H(j)}\right) - \left(w_{H(j)} + w_{L(j)}\right) - p_a(j)\left(a_{H(j)} + a_{L(j)}\right)$$

Since revenues depend on workers' joint productivity, $f\left(0, \alpha_{H(j)}\right) = f\left(\alpha_{L(j)}, 0\right) = 0$. Therefore, whenever a worker rejects the offer, the firm exits and both workers receive their respective outside options, u_i.

The environment, as it is, admits a feasible, closed-form equilibrium. I introduce four additional assumptions that furhter discipline the framework's empirical predictions around the wage-amenities tradeoff.

Assumption 1. *Worker productivity α_i is fixed.*

The assumption shuts down the productivity channel. That is, firms accommodate higher wage floors by exhorting greater effort from workers who earn the statutory minimum (Board and Meyer-Ter-Vehn, 2015; Wang, 2025). On paper, it keeps amenities one-dimensional; allowing for endogenous productivity would introduce another dimension related to effort costs. At the same time, it seals similar margins for adjustment, such as pass-throughs to prices (Allegretto and Reich, 2018; Harasztosi and Lindner, 2019; Renkin et al., 2022).

Assumption 2. *Firms earn strictly positive profits, even when paying the minimum wage to all workers.*

Wage floors raise total labor costs, incentivizing firms in the framework to exit the market in response. The assumption rules out exit by ensuring that firms can always accommodate higher wages; a reasonable restriction given prior evidence of muted employment declines. Recent empirical evidence from the United States and Germany points to monopsony power, with firms adjusting markdowns rather than employment (Azar et al., 2024; Berger et al., 2025; Faia et al., 2026). In the context of Brazil, Dobbin et al. (2025) report significant markdowns among large, more productive firms, which provides some support for the assumption.

Before moving on to the next two assumptions, I classify firms into three categories based on their initial exposure to the binding minimum wage, regardless of their average wage level.

Definition (Firm categories)**.** *I define a high-paying employer $\mathcal{H}$ as the firm that pays wages strictly above the minimum wage to all its employees. A medium-paying employer, instead, is the firm in which only the low-productivity worker earns the statutory minimum. In addition, I split medium-paying firms into two distinct categories. On the one hand, specialized employers $\mathcal{S}$ exhibit lower workplace heterogeneity, defined by the gap between u_H and u_L. On the other hand, vertical employers $\mathcal{V}$ employ a more heterogeneous workforce. Finally, I define a low-paying employer $\mathcal{L}$ as the firm that pays the minimum wage to all its employees. In addition, I define $\pi_{\mathcal{X}}$ as the share of firms in a given category $\mathcal{X}$, where $\mathcal{X} \in \{\mathcal{H}, \mathcal{S}, \mathcal{V}, \mathcal{L}\}$. By construction, $\mathcal{L} \cup \mathcal{S} \cup \mathcal{V} \cup \mathcal{H} = \mathcal{J}$, and $\pi_{\mathcal{H}} + \pi_{\mathcal{S}} + \pi_{\mathcal{V}} + \pi_{\mathcal{L}} = 1$.*

Assumption 3. *The joint distribution of the unitary price of amenities p_a and workers' outside options u_L and u_H is atomless.*

In other words, no two firms simultaneously share the same firm-specific unitary price of job amenities and the same workers' outside options. Under the partial equilibrium assumption, these variables are taken as given and fully capture heterogeneity across firms.

Although optimal contracts ultimately depend on the firm's category, this classification is based solely on these three exogenous characteristics. As a result, there are no incentives for bunching. That is, no endogenous choice in the model can generate mass points at any particular combination of these parameters. In any case, the assumption works as a convenience device, ensuring that firms located at the margin between categories cannot disproportionately drive the results.

Assumption 4. *Workplace heterogeneity at specialized firms $\mathcal{S}$—defined as the outside options ratio between high-paid and low-paid workers—is bounded by 2^{γ}.*

$$\frac{u_{H(s)}}{u_{L(s)}} < 2^{\gamma} \quad \forall s \in \mathcal{S}$$

The assumption places an upper bound on workplace heterogeneity, ultimately disciplining within-firm wage differentials at specialized firms. In the empirical analysis, it serves as a sufficient but not necessary condition and therefore can be relaxed. Nonetheless, I return to it in Section (A.4), where I derive a prediction that matches the empirical observation that wage inequality decreases following increases in the minimum wage.

Assumption 5. *The outside options of high-paid workers employed at low-paying firms and of low-paid workers employed at specialized firms exhibit a sufficiently low dispersion. More specifically, I assume that*

$$\pi_{\mathcal{L}} \, \mathbb{V}_{\mathcal{L}}\Big[u_H\Big] + \pi_{\mathcal{S}} \, \mathbb{V}_{\mathcal{S}}\Big[u_L\Big]$$

is relatively small.

Intuitively, the assumption imposes that workers earning the minimum wage across firms in the same category do not exhibit large differences. As before, the condition acts as a sufficient but not necessary condition in the empirical analysis. Consequently, I discuss it further in Section A.4, when I derive a prediction about job amenities dispersion following increases in the minimum wage.

Worker reallocations. Engbom and Moser (2022) offer an alternative interpretation of the limited employment effects in Brazil: workers reallocate to more productive firms. Although a static framework cannot capture these dynamics, a small tweak to the model can align its interpretation with the author's findings. Let the incumbent firm exit before contracts are posted, and let an entrant firm replace it and post its own offers. Moreover, let the entrant face a $p_a(j)$ sufficiently low to break even. Since optimal wages and job amenities for constrained firms do not depend on $p_a(j)$, both versions are isomorphic. More importantly, empirical predictions stay the same as long as entrant and exiting firms belong to the same firm category, which holds by construction.

Single firm's problem. For convenience, I drop the firm index j from wages w_i, job amenities a_i, outside options u_i, and the unitary cost p_a when discussing the single firm's problem. Given Assumption (1), optimal wages and job amenities arise from the firm's cost minimization problem, subject to each worker's participation and minimum-wage constraints.

$$\begin{aligned}
& \min_{\{(w_i,a_i)\}_{i=L}^{H}} && (w_L + w_H) + p_a(a_L + a_H) \\
(PC) \qquad & \text{s.t.} && w_i^{\gamma} a_i \;\geqslant\; u_i, \qquad i \in \{L, H\} \\
(MW) \qquad & && w_i \;\geqslant\; \underline{w}, \qquad i \in \{L, H\}
\end{aligned}$$

I can solve the problem using the Lagrangian approach. Let λ_i denote the multiplier on (PC_i) and μ_i the multiplier on (MW_i), so that

$$\begin{aligned}
\mathcal{L} \;=\; & (w_L + w_H) + p_a(a_L + a_H) \\
& + \; \lambda_L(u_L - w_L^{\gamma} a_L) \;+\; \mu_L(\underline{w} - w_L) \\
& + \; \lambda_H(u_H - w_H^{\gamma} a_H) + \; \mu_H(\underline{w} - w_H)
\end{aligned}$$

with the following first-order conditions

$$\begin{aligned}
(w_H) \qquad & 1 - \lambda_H \gamma w_H^{\gamma-1} a_H - \mu_H = 0 \\
(a_H) \qquad & p_a - \lambda_H w_H^{\gamma} = 0 \\
(\lambda_H) \qquad & \lambda_H \left(u_H - w_H^{\gamma} a_H\right) = 0 \\
& \lambda_H \geqslant 0 \\
& u_H - w_H^{\gamma} a_H \leqslant 0
\end{aligned}
\qquad\qquad
\begin{aligned}
(w_L) \qquad & 1 - \lambda_L \gamma w_L^{\gamma-1} a_L - \mu_L = 0 \\
(a_L) \qquad & p_a - \lambda_L w_L^{\gamma} = 0 \\
(\lambda_L) \qquad & \lambda_L \left(u_L - w_L^{\gamma} a_L\right) = 0 \\
& \lambda_L \geqslant 0 \\
& u_L - w_L^{\gamma} a_L \leqslant 0
\end{aligned}$$

$$(\mu_H) \qquad \begin{aligned} \mu_H\left(\underline{w} - w_H\right) &= 0 \\ \mu_H &\geqslant 0 \\ \underline{w} - w_H &\leqslant 0 \end{aligned} \qquad (\mu_L) \qquad \begin{aligned} \mu_L\left(\underline{w} - w_L\right) &= 0 \\ \mu_L &\geqslant 0 \\ \underline{w} - w_L &\leqslant 0 \end{aligned}$$

In the following subsections, I derive the solutions for two separate situations. First, when firms can tailor job amenities specifically to each worker, $a_L \neq a_H$. Second, when firms must commit to common levels of job amenities, $a_L = a_H = a$.

2. Flexible amenities

With worker-specific amenities, each contract can be solved separately. Equation (a_i) determines that $\lambda_i^* = p_a w_i^{-\gamma} > 0$, which implies that participation constraints are always binding and

$$a_i^* = \frac{u_i}{w_i^{*\gamma}}$$

Replacing λ_i^* into equation (w_i) determines that $\mu_i^* = 1 - p_a \gamma a_i w_i^{-1} \geqslant 0$, so optimal contracts depend on whether the minimum wage condition binds for the worker.

Definition ($\underline{u}$)**.** *Let $\underline{u} \equiv \underline{w}^{1+\gamma}\left(\gamma p_a\right)^{-1}$ denote the threshold that determines whether the firm faces a binding (MW_i).*

The threshold $\underline{u}$ maps a positive μ_i^* onto a condition based on the relationship between the worker's outside option and the minimum wage. Namely, employers pay the minimum wage to workers with $u_i \leqslant \underline{u}$.

Notice that $\underline{u}$ is a firm-specific threshold, emphasizing that firms with access to cheaper job amenities (lower p_a) pay lower wages and are, everything else equal, more exposed to changes in the minimum wage.[13]

Lemma 1. *When $u_i \geqslant \underline{u}$, optimal contracts offer workers*

$$w^* = u_i^{\frac{1}{1+\gamma}}\left(p_a \gamma\right)^{\frac{1}{1+\gamma}}$$

and job amenities

$$a^* = u_i^{\frac{1}{1+\gamma}}\left(p_a \gamma\right)^{\frac{-\gamma}{1+\gamma}}$$

Proof. Combining $\mu_i^* = 0$ and a_i^* suffice to determine optimal contracts. □

Notice that the worker's utility has a Cobb-Douglas functional form, so that θ denotes the share of compensation offered as amenities.

Comparative statics portray intuitive results. Both w_i and a_i are increasing and concave in u_i, consistent with empirical work showing a positive correlation between wages and job amenities (Sockin, 2022; Maestas et al., 2023). On the contrary, w_i is increasing and concave in p_a, while a_i is decreasing and convex, reflecting how firms substitute the relatively expensive form of compensation for the cheapest one.

Lemma 2. *When $u_i < \underline{u}$, optimal contracts offer the worker the minimum wage $\underline{w}$ and job amenities $a_i^* = u_i \underline{w}^{-\gamma}$.*

[13] The claim may appear counterintuitive: more productive firms plausibly face lower costs of providing amenities and pay the minimum wage to fewer workers. The reasoning, however, overlooks that the opportunity cost of relaxing working conditions declines with productivity. Consistent with this mechanism, Humlum et al. (2025) show that high-paying firms offer jobs with better perks but also greater stress, turnover, and related workplace disadvantages.

Proof. Both a_i^* and $w_i^* = \underline{w}$ immediately reveals the optimal solution. □

Thus, a_i is decreasing and convex in $\underline{w}$, implying that higher wage floors lead to greater reductions in job amenities. Though a_i is still increasing in u_i, the relationship is no longer concave but linear, indicating that firms reduce job amenities among low-productivity workers more aggressively in the presence of wage floors. Most notably, a_i no longer depends on p_a, since job amenities for constrained workers become pinned down by (PC_i).

Labor costs. Assumption 2 eliminates the need to impose a firm participation constraint. Even so, the framework captures how amenities can attenuate layoffs following minimum wage increases. I take the difference in amenity expenditures across the two cases

$$\Delta LC = p_a u_i \left((p_a u_i \gamma)^{\frac{-\gamma}{1+\gamma}} - \underline{w}^{-\gamma} \right)$$

so that ΔLC denotes labor cost savings relative to a setting with fixed amenities.

The result shows that the mechanism strengthens as wage floors rise, since ΔLC increases with $\underline{w}$. Moreover, ΔLC is non-monotonic in p_a and u_i. Savings increase for $u_i < \underline{u}(1+\gamma)^{-(1+\gamma)}$ and decrease thereafter. Therefore, amenity-driven savings concentrate among low-productivity workers, especially when the unit cost of amenities is higher.

Thus far, the setting with flexible amenities has helped to clarify the mechanisms and the intuition behind the framework. The analysis next considers a setting empirically closer to Brazil, in which firms cannot differentiate amenities across workers. Before proceeding, note that wages increase in u_i and, by construction, in α_i. In consequence, I henceforth refer to low- and high-paid workers rather than low- and high-productivity workers, which better aligns with the intuition behind the empirical predictions.

3. Firm-level amenities

With firm-specific amenities, equations (a_L) and (a_H) collapse into a unique first order condition

$$(a) \qquad 2p_a - \lambda_L w_L^{\gamma} - \lambda_H w_H^{\gamma} = 0$$

Unlike the previous case, contracts must be solved jointly, leading to four solutions of the program. Either way, the participation constraint for the high-paid worker remains always binding.

Lemma 3. *(PC_H) is always binding.*

Proof. Let (PC_H) be slack and replace $\lambda_H = 0$ in equations (w_h) and (a). The first one yields $\mu_H^* = 1$, which immediately implies that $a^* > u_H \underline{w}^{-\gamma}$. The second one determines that λ_L is positive, so $u_L = w_L^{\gamma} a^*$. Given that $w_L \geqslant \underline{w}$ and $u_H > u_L$ by construction, combining both statements leads to the following contradiction

$$u_L > \left(\frac{w_L}{\underline{w}} \right)^{\gamma} u_H$$

so that (PC_H) has to be binding. □

The firm selects the optimal contract from four characterizations based on its exposure to the wage floor, which, in turn, depends on workers' outside options and the unit cost of amenities. For notational convenience, let me define $P_a \equiv (2p_a\gamma)^{\frac{1}{1+\gamma}}$, which denotes a renormalization of the firm's amenity price.

I map each characterization to a distinct firm category to provide better intuition into the firm's choices. Each employer then faces three firm-specific thresholds relative to the minimum wage that identify the optimal contract. As the minimum wage increases and surpasses each threshold, the bindingness of the wage floor strengthens, and the firm adjusts its response to further increases.

The first threshold, ω_1, classifies the firm as either a high-paying or a specialized employer. In the former case, the firm pays wages above the statutory minimum and is thus unaffected by further marginal increases. In the latter, instead, the firm employs workers with small differences in their outside options but pays the minimum wage to the low-paid employee. By having a less heterogeneous workforce, the specialized firm prefers to pass along the minimum wage increase to the high-paid worker, even though she earns above the statutory minimum.

Definition (ω_1)**.** *Let*

$$\omega_1 \equiv P_a u_L^{\frac{1}{\gamma}} \left(u_L^{\frac{1}{\gamma}} + u_H^{\frac{1}{\gamma}} \right)^{-\frac{1}{1+\gamma}}$$

denote the threshold that separates high-paying ($\mathcal{H}$) from specialized firms ($\mathcal{S}$).

Three things worth mentioning about this threshold. First, P_a relaxes the restriction by making amenities relatively more expensive, thereby pushing the firm to offer higher wages to everyone in the first place. Second, a more heterogeneous workforce tightens the restriction by pushing down the wage of the low-paid worker. Third, a higher outside option for the low-paid worker softens the constraint. Naturally, a higher u_L, by construction, implies that the firm faces a more homogeneous worker composition, though a simultaneous increase in both outside options would keep wages away from the statutory minimum.

The second threshold, ω_2, distinguishes a medium-paying firm as having either a specialized or a vertical structure. The difference between the two is that a vertical employer has a more heterogeneous workforce, i.e., workers exhibit a higher gap in outside options. As a result, the firm prefers to absorb the minimum wage increase, and the low-paid worker pockets a differential.

Definition (ω_2)**.** *Let*

$$\omega_2 \equiv P_a u_L^{\frac{1}{\gamma}} \left(u_H^{\frac{1}{\gamma}} \right)^{-\frac{1}{1+\gamma}}$$

denote the threshold that separates specialized ($\mathcal{S}$) from vertical firms ($\mathcal{V}$).

As before, P_a increases the threshold by raising the cost of amenities relative to wages. Workforce heterogeneity also tightens the restriction. In particular, because compensating differentials are costly in this scenario, it is more profitable for the firm to absorb the wage floor that affects the low-paid worker.

The last threshold, ω_3, determines whether the firm falls under the definition of a vertical or low-paying employer, where the latter pays the minimum wage to both workers.

Definition (ω_3)**.** *Let*

$$\omega_3 \equiv P_a u_H^{\frac{1}{1+\gamma}}$$

denote the threshold that separates vertical ($\mathcal{V}$) from low-paying firms ($\mathcal{L}$).

Two things worth mentioning here. First, ω_3 does not depend on u_L. By construction, the firm cannot pay a lower wage to the high-paid worker, so this condition suffices. Second, as workforce heterogeneity falls, $\omega_2 \to \omega_3$. Therefore, the bindingness of the

wage floor depends solely on the unit cost of job amenities and the outside option of the matched team.

Proposition 1 (High-paying firms)**.** *When $\underline{w} < \omega_1$, the optimal contract offers workers firm-specific job amenities equal to*

$$a^* = P_a^{-\gamma} \left(u_L^{\frac{1}{\gamma}} + u_H^{\frac{1}{\gamma}} \right)^{\frac{\gamma}{1+\gamma}}$$

and worker-specific wages equal to

$$w_i^* = P_a u_i^{\frac{1}{\gamma}} \left(u_L^{\frac{1}{\gamma}} + u_H^{\frac{1}{\gamma}} \right)^{-\frac{1}{1+\gamma}}$$

Proof. By definition, high-paying firms have $\mu_L = \mu_H = 0$. Replacing these values in equations (w_H) and (w_L) shows that λ_L and λ_H are positive, further implying that (PC_L) also binds.

Therefore, I can isolate w_i on each (PC_i) to replace these expressions in λ_i, which results in the following expression

$$\lambda_i = u_i^{\frac{1-\gamma}{\gamma}} \left(\gamma a^{\frac{1}{\gamma}} \right)^{-1}$$

that I can replace in equation (a).

Isolating a yields the optimal level of job amenities; thus, I can obtain optimal worker-specific wages w_i^* by replacing a^* in each (PC_i).

Finally, I need to check that the threshold's condition holds. Notice that the expression for ω_1 coincides with w_L, so by construction, $w_H > w_L = \omega_1 > \underline{w}$. □

The comparative statics reveal interesting interplays in compensation. As with Lemma (1), a is decreasing and convex in p_a, whereas wages are increasing and concave. Job amenities increase with both u_L and u_H, suggesting that the firm sets a to target an average level of compensation and uses wages to satisfy participation constraints. In fact, wages w_i are increasing and concave in u_i, but decrease with u_{-i}. The condition $\gamma \geqslant 1$, which implies that workers do not have stronger preferences for amenities, is sufficient for job amenities to be concave in outside options and for wages to be convex in the coworker's outside option. Otherwise, the sign of the second derivative is indeterminate.

Proposition 2 (Medium-paying firms)**.** *When $\omega_1 \leqslant \underline{w} < \omega_3$, the firm sets the low-paid worker's wage at the minimum wage. If $\underline{w} < \omega_2$, the firm serves as a specialized employer and offers a compensating differential to the high-paid worker. Then, the wage of the high-paid worker amounts to*

$$w_H^* = \underline{w} \left(\frac{u_H}{u_L} \right)^{1/\gamma}$$

and the optimal contract offers firm-specific job amenities equal to

$$a^* = u_L \underline{w}^{-\gamma}$$

On the contrary, when $\omega_2 \leqslant \underline{w}$, the firm falls under the definition of a vertical employer.

Therefore, the optimal contract offers job amenities equal to

$$a^* = P_a^{-\gamma} u_H^{\frac{1}{1+\gamma}}$$

and the following wage to the high-paid worker

$$w_H^* = P_a u_H^{\frac{1}{1+\gamma}}$$

Proof. By definition, medium-paying firms have $\mu_H = 0$ and $\mu_L > 0$. By using the same expression for λ_H as in Proposition (1) and substituting it into equation (a), I can obtain the following restriction for λ_L

$$\lambda_L = \frac{P_a^{1+\gamma} - u_H^{\frac{1}{\gamma}} a^{-\frac{1+\gamma}{\gamma}}}{\gamma \underline{w}^{\gamma}} \geqslant 0$$

Therefore, there are two potential characterizations for optimal contracts. Let me assume that λ_L is positive. Then, it immediately follows that (PC_L) is binding, which immediately pins down $a^* = \underline{w}^{-\gamma} u_L$. Replacing the optimal level of job amenities in (PC_H) thereby yields the optimal wage for the high-paid worker, completing the characterization of the specialized firm's optimal contract.

Next, I assume that $\lambda_L = 0$, which immediately reveals the optimal level of job amenities for the vertical firm. In turn, I can replace a^* in (PC_H) and obtain w_H^*.

Lastly, I need to check that all thresholds' conditions hold. First, $\omega_1 \leqslant \underline{w}$ trivially follows from $\mu_L > 0$. Second, a specialized firm pays a w_H^* that lies strictly above the minimum wage and, in the case of a vertical employer, the high-paid worker's wage coincides with ω_3. Therefore, $\underline{w} < \omega_3$ holds by construction. Third, $\lambda_L = 0$ implies that

$$U_L > \underline{w}^{\gamma} P_a^{-\gamma} u_H^{\frac{1}{1+\gamma}} > u_L \implies \underline{w} > P_a u_L^{\frac{1}{\gamma}} \left(u_H^{\frac{1}{\gamma}} \right)^{-\frac{1}{1+\gamma}} = \omega_2$$

Therefore, the specialized firm's characterization of the optimal contract holds when $\underline{w} \leqslant \omega_2$, while the characterization for the vertical employer holds otherwise. □

A specialized firm no longer uses job amenities to target an average level of compensation, but to satisfy the participation constraint of the low-paid worker. As a result, a increases linearly only in u_L and decreases convexly in $\underline{w}$. Regarding the wage of the high-paid worker, it increases linearly in $\underline{w}$ and, when $\gamma > 1$, it increases concavely in u_H, while decreasing convexly in u_L. Otherwise, the signs of the second derivatives become indeterminate. In conclusion, setting a lower a due to the wage floor pushes the firm to offer a compensating differential to the high-paid worker, which explains the upward spillovers in the wage distribution.

However, a vertical employer acts differently. The firm sets job amenities to satisfy the participation constraint of the high-paid worker, allowing the low-paid worker to keep a wage differential. As a result, the optimal contract offered to the high-paid worker no longer depends on u_L. Comparative statics reveal that a is decreasing and convex in p_a while increasing and concave in u_H. w_H, on the other hand, is increasing and concave in both p_a and u_H.

These results have two implications. Spillovers arise and grow with workplace heterogeneity, defined as the gap between u_H and u_L. However, too much heterogeneity dissipates the effect, since the firm weighs compensating differentials as too costly and thus prefers to directly absorb the minimum wage increase.

Proposition 3 (Low-paying firms)**.** *When $\omega_3 \leqslant \underline{w}$, the optimal contract offers both workers the minimum wage $\underline{w}$ and firm-specific job amenities equal to*

$$a = u_H \underline{w}^{-\gamma}$$

Proof. By definition, a low-paying firm has $\mu_H > 0$ and $\mu_L > 0$. Therefore, (PC_H) pins down a^*. Furthermore, the threshold's condition holds from Proposition (2). □

Compared to a specialized firm, a increases linearly in u_H but still decreases convexly in $\underline{w}$. Compared to the vertical employer, the firm still uses job amenities to address the participation constraint of the high-paid worker. Thus, (PC_L) remains slack, and the low-paid worker pockets the full wage differential, $u_H - u_L$.

4. Empirical predictions

The analysis henceforth imposes firm-specific job amenities and uses results in Propositions (1) to (3). The subsection illustrates that empirical predictions for an increase in the minimum wage may not be trivial when firms cannot tailor job amenities to workers. In particular, wage inequality can increase, and disparities in job amenities can fall. The underlying reason connects to the third and most important prediction of the model: some firms may pass along the minimum wage increase to higher earners. Next, I present testable implications of the model and show sufficient conditions under which the predictions align directly with the empirical patterns observed in the Brazilian context.

Lemma 4. *Let $\pi_{\mathcal{X}}(\underline{w})$ denote the share of firms in category $\mathcal{X}$, with $\mathcal{X} \in \{\mathcal{L}, \mathcal{V}, \mathcal{S}, \mathcal{H}\}$. Then, changes in the share of firms in each category following a minimum wage increase are negligible. That is,*

$$\frac{\partial \pi_{\mathcal{X}}(\underline{w})}{\partial \underline{w}} \approx 0$$

Proof. Without loss of generality, consider the share of specialized firms $\pi_{\mathcal{S}}(\underline{w})$. Proposition (2) shows that the share depends on the thresholds ω_1 and ω_2, such that

$$\pi_{\mathcal{S}}(\underline{w}) = F_2(\underline{w}) - F_1(\underline{w})$$

where F_1 and F_2 represent the cumulative distribution functions of $\omega_1(p_a, u_L, u_H)$ and $\omega_2(p_a, u_L, u_H)$ across firms, respectively.

By taking derivatives with respect to the minimum wage $\underline{w}$, I obtain the following expression

$$\frac{\partial \pi_{\mathcal{S}}(\underline{w})}{\partial \underline{w}} = f_2(\underline{w}) - f_1(\underline{w}) \approx 0$$

where f_1 and f_2 correspond to the probability density functions, and are negligible at any point due to Assumption (3). □

Lemma (4) serves as a convenience device to eliminate the need to introduce further assumptions about the distribution of firms across categories. Since the thresholds defined in Subsection (A.3) do not depend on endogenous choices, the assumption that firms do not bunch around them, nor any other point in the distribution, appears reasonable.

In any case, none of the analytical results that follow depend on the exact magnitudes of any $\pi_{\mathcal{X}}$. Introducing small perturbations to the shares or, equivalently, small buffers to include thresholds, leaves all results unchanged, further reinforcing that neglecting changes in shares is inconsequential.

Lemma 5. *Let* $\mathbb{E}[w] \geqslant 2\underline{w}$. *Then,*

$$\mathbb{V}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right] + \mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right]\left(\mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right] - \frac{\mathbb{E}[w]}{\underline{w}}\right) \leqslant 0$$

Proof. Define

$$A \equiv \left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}} \qquad \text{and} \qquad B \equiv P_a\, u_L^{\frac{1}{1+\gamma}} \underline{w}^{-1}$$

Proposition (2) states that specialized firms $\mathcal{S}$ satisfy $\underline{w} \in [\omega_1, \omega_2]$. By rearranging terms in ω_1 and ω_2, I can define the following interval for A

$$A \in [L(A), U(A)]$$

where $L(A) = B\,A\,(1+A)^{-\frac{1}{1+\gamma}}$ and $U(A) = B\,A^{\frac{\gamma}{1+\gamma}}$. Next, let me define

$$T_{\mathcal{S}}(A) = \mathbb{V}_{\mathcal{S}}[A] + \mathbb{E}_{\mathcal{S}}[A]\left(\mathbb{E}_{\mathcal{S}}[A] - \frac{\mathbb{E}[w]}{\underline{w}}\right)$$

and

$$\hat{A} \equiv \underset{a \in [L(A), U(A)]}{\operatorname{argmax}} \; T_{\mathcal{S}}(a)$$

By construction, I can set the following inequality

$$\begin{aligned} T_{\mathcal{S}}(A) &\leqslant \mathbb{V}_{\mathcal{S}}\left[\hat{A}\right] + \mathbb{E}_{\mathcal{S}}\left[\hat{A}\right]\left(\mathbb{E}_{\mathcal{S}}\left[\hat{A}\right] - \frac{\mathbb{E}[w]}{\underline{w}}\right) \\ &\leqslant \mathbb{E}_{\mathcal{S}}\left[\hat{A}^2\right] - \mathbb{E}_{\mathcal{S}}\left[\hat{A}\right]^2 + \mathbb{E}_{\mathcal{S}}\left[\hat{A}\right]^2 - \mathbb{E}_{\mathcal{S}}\left[\hat{A}\right]\frac{\mathbb{E}[w]}{\underline{w}} \\ &\leqslant \mathbb{E}_{\mathcal{S}}\left[\hat{A}\right]\underbrace{\left(E_{\mathcal{S}}\left[\hat{A}\right] - \frac{\mathbb{E}[w]}{\underline{w}}\right)}_{\text{negative}} \end{aligned}$$

which yields that $T_{\mathcal{S}}$ is negative under Assumption (4) and that the average wage is at least twice the minimum wage.

$$E_{\mathcal{S}}\left[\hat{A}\right]^{\gamma} < 2^{\gamma} \qquad \text{and} \qquad \frac{\mathbb{E}[w]}{\underline{w}} \geqslant 2$$

□

Lemma (5) follows from the combination of $\mathbb{E}[w] \geqslant 2\underline{w}$ and Assumption (4). The former condition stems from the observed ratio of the average to the minimum hourly wage in Brazil. In particular, the ratio ranged from 2.05 (in 2016) to 2.11 (in 2014), which backs this condition as suitable.

Assumption (4), on the other hand, is a sufficient condition that I impose to prevent the minimum wage from skewing the wage distribution to the right. In the framework, workplace heterogeneity maps one-to-one with within-firm wage differentials among specialized firms, so both interpretations of the assumption are interchangeable. Intuitively, Proposition (2) shows that wages in specialized firms increase linearly in $\underline{w}$,

though at different rates. The assumption places an upper bound on the growth rate of higher earners among these types of firms, so it cannot exceed the ratio of the average to the minimum hourly wage. Otherwise, wage compression could occur solely at the upper tail, leaving a wider lower-tail wage dispersion.

Proposition 4 (Wage compression)**.** *Wage dispersion* $\mathbb{V}[w]$ *is not monotonic in* $\underline{w}$*. Assumption (4) is a sufficient condition for wage dispersion to decrease with* $\underline{w}$*.*

Proof. Using Propositions (1) to (3), I can express the first and second moments of w as

$$\mathbb{E}[w] = \pi_{\mathcal{L}}\underline{w} + 1/2\pi_{\mathcal{V}}\Big(\underline{w} + \mathbb{E}_{\mathcal{V}}[w_H]\Big) + 1/2\pi_{\mathcal{S}}\underline{w}\left(1 + \mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right]\right) + \pi_{\mathcal{H}}\mathbb{E}_{\mathcal{H}}[w]$$

and

$$\mathbb{E}\left[w^2\right] = \pi_{\mathcal{L}}\underline{w}^2 + 1/2\pi_{\mathcal{V}}\Big(\underline{w}^2 + \mathbb{E}_{\mathcal{V}}\left[w_H^2\right]\Big) + 1/2\pi_{\mathcal{S}}\underline{w}^2\left(1 + \mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{2}{\gamma}}\right]\right) + \pi_{\mathcal{H}}\mathbb{E}_{\mathcal{H}}\left[w^2\right]$$

where $\mathbb{E}_{\mathcal{X}}[\cdot]$ highlights heterogeneities in wages and workplace composition within firms in category $\mathcal{X}$. Next, I can take first derivatives of the previous expressions with respect to $\underline{w}$ and get

$$\frac{\partial\mathbb{E}[w]}{\partial\underline{w}} = 1/2\,(2\,\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}) + 1/2\,\pi_{\mathcal{S}}\,\mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right]$$

and

$$\frac{\partial\mathbb{E}\left[w^2\right]}{\partial\underline{w}} = \underline{w}\,(2\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}) + \underline{w}\,\pi_{\mathcal{S}}\,\mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{2}{\gamma}}\right]$$

where I employ Lemma (4) to ignore the derivatives of the share of firms across categories with respect to the minimum wage. Next, let me write the second centered moment of w as

$$\mathbb{V}[w] = \mathbb{E}\left[w^2\right] - \mathbb{E}[w]^2$$

and take derivatives with respect to $\underline{w}$. Thus,

$$\begin{aligned}\frac{\partial\mathbb{V}[w]}{\partial\underline{w}} &= \frac{\partial\mathbb{E}\left[w^2\right]}{\partial\underline{w}} - 2\mathbb{E}[w]\frac{\partial\mathbb{E}[w]}{\partial\underline{w}} \\ &= \Big(\underline{w} - \mathbb{E}[w]\Big)\Big(2\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}\Big) \\ &\quad + \pi_{\mathcal{S}}\left(\underline{w}\mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{2}{\gamma}}\right] - \mathbb{E}[w]\,\mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right]\right)\end{aligned}$$

By adding and substracting

$$\underline{w}\,\pi_{\mathcal{S}}\,\mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right]^2$$

and rearranging the second term, I obtain

$$\frac{\partial \mathbb{V}[w]}{\partial \underline{w}} = \underbrace{\Big(\underline{w} - \mathbb{E}[w]\Big)}_{\text{negative}} \Big(2\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}\Big) + \pi_{\mathcal{S}}\, \underline{w} \underbrace{\left(\mathbb{V}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right] + \mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right]\left(\mathbb{E}_{\mathcal{S}}\left[\left(\frac{u_H}{u_L}\right)^{\frac{1}{\gamma}}\right] - \frac{\mathbb{E}[w]}{\underline{w}}\right)\right)}_{\text{Lemma (5)}}$$

which is strictly negative by Lemma (5). □

The second term in Proposition (4)illustrates how spillovers across the wage distribution make the prediction about wage inequality less obvious. Once $\underline{w}$ approaches the average wage, increases in the federal minimum wage can increase the skewness of the wage distribution.

Notice that Lemma (5) undoes a step in the algebra of Proposition (2). However, the second term in the latter sheds light on the counter-effects of increasing the minimum wage when there are significant spillovers. On the one hand, the wages of high-paid workers grow faster than the minimum wage, so higher wage floors increase wage dispersion within specialized firms. On the other hand, when spillovers affect earners above the average wage, wage disparities across categories can increase significantly, leading to a wider lower-tail wage dispersion.

Assumption (4) serves as a sufficient condition for the model to match the decline in wage inequality over time in Brazil. So the faster wage growth among high-paid workers does not translate directly into a more unequal wage distribution. Even if the sufficient condition is not met, the empirical evidence suggests that the decline in wage variance across categories has absorbed the increase within specialized firms. However, the steady pass-through of minimum wage increases across the whole wage distribution should raise awareness regarding potential pitfalls in the policy's effectiveness. In particular, in local labor markets with relatively lower average wages.

Proposition 5 (Amenities dispersion)**.** *Job amenities dispersion* $\mathbb{V}[a]$ *is not monotonic in* $\underline{w}$*. Furthermore, Assumption (5) is a sufficient condition for amenities dispersion to increase with* $\underline{w}$*.*

Proof. Using Propositions (1) to (3), I can write the first and second moment of a as

$$\mathbb{E}[a] = \pi_{\mathcal{L}} \underline{w}^{-\gamma} \mathbb{E}_{\mathcal{L}}[u_H] + \pi_{\mathcal{V}} \mathbb{E}_{\mathcal{V}}[a] + \pi_{\mathcal{S}} \underline{w}^{-\gamma} \mathbb{E}_{\mathcal{L}}[u_L] + \pi_{\mathcal{H}} \mathbb{E}_{\mathcal{H}}[a]$$

and

$$\mathbb{E}\left[a^2\right] = \pi_{\mathcal{L}} \underline{w}^{-2\gamma} \mathbb{E}_{\mathcal{L}}\left[u_H^2\right] + \pi_{\mathcal{V}} \mathbb{E}_{\mathcal{V}}\left[a^2\right] + \pi_{\mathcal{S}} \underline{w}^{-2\gamma} \mathbb{E}_{\mathcal{L}}\left[u_L^2\right] + \pi_{\mathcal{H}} \mathbb{E}_{\mathcal{H}}\left[a^2\right]$$

Thus, I can take first derivatives of the previous expressions with respect to $\underline{w}$ and get

$$\frac{\partial \mathbb{E}[a]}{\partial \underline{w}} = -\gamma\, \underline{w}^{-(\gamma+1)} \Big(\pi_{\mathcal{L}}\, \mathbb{E}_{\mathcal{L}}[u_H] + \pi_{\mathcal{S}}\, \mathbb{E}_{\mathcal{S}}[u_L]\Big)$$

and

$$\frac{\partial \mathbb{E}[a]}{\partial \underline{w}} = -2\gamma\, \underline{w}^{-(2\gamma+1)} \Big(\pi_{\mathcal{L}}\, \mathbb{E}_{\mathcal{L}}\left[u_H^2\right] + \pi_{\mathcal{S}}\, \mathbb{E}_{\mathcal{S}}\left[u_L^2\right]\Big)$$

where I employ Lemma (4) to dismiss the derivatives of the share of firms across the categories with respect to the minimum wage. Then, the second centered moment of a

can be written as

$$\mathbb{V}[a] = \mathbb{E}\left[a^2\right] - \mathbb{E}[a]^2$$

with a first derivative with respect to $\underline{w}$, equal

$$\begin{aligned}
\frac{\partial \mathbb{V}[a]}{\partial \underline{w}} &= \frac{\partial \mathbb{E}\left[a^2\right]}{\partial \underline{w}} - 2\,\mathbb{E}[a]\,\frac{\partial \mathbb{E}[a]}{\partial \underline{w}} \\
&= -2\gamma\,\underline{w}^{-(2\gamma+1)}\Big(\pi_{\mathcal{L}}\mathbb{E}_{\mathcal{L}}\left[u_H^2\right] + \pi_{\mathcal{S}}\mathbb{E}_{\mathcal{S}}\left[u_L^2\right]\Big) \\
&\quad + 2\gamma\,\underline{w}^{-(\gamma+1)}\,\mathbb{E}[a]\Big(\pi_{\mathcal{L}}\mathbb{E}_{\mathcal{L}}[u_H] + \pi_{\mathcal{S}}\mathbb{E}_{\mathcal{S}}[u_L]\Big)
\end{aligned}$$

By adding and substracting

$$2\,\gamma\,\underline{w}^{-(2\gamma+1)}\pi_{\mathcal{L}}\,\mathbb{E}_{\mathcal{L}}[u_H]^2 \quad \text{and} \quad 2\,\gamma\,\underline{w}^{-(2\gamma+1)}\pi_{\mathcal{S}}\,\mathbb{E}_{\mathcal{S}}[u_L]^2$$

and rearranging the first term, I obtain

$$\begin{aligned}
\frac{\partial \mathbb{V}[a]}{\partial \underline{w}} &= -2\gamma\,\underline{w}^{-(2\gamma+1)}\underbrace{\Big(\pi_{\mathcal{L}}\mathbb{V}_{\mathcal{L}}[u_H] + \pi_{\mathcal{S}}\mathbb{V}_{\mathcal{S}}[u_L]\Big)}_{\text{Assumption (5)}} \\
&\quad + 2\gamma\,\underline{w}^{-(\gamma+1)}\,\pi_{\mathcal{L}}\mathbb{E}_{\mathcal{L}}[u_H]\underbrace{\Big(\mathbb{E}[a] - \mathbb{E}_{\mathcal{L}}[a]\Big)}_{\text{positive}} \\
&\quad + 2\gamma\,\underline{w}^{-(\gamma+1)}\,\pi_{\mathcal{S}}\mathbb{E}_{\mathcal{S}}[u_L]\underbrace{\Big(\mathbb{E}[a] - \mathbb{E}_{\mathcal{S}}[a]\Big)}_{\text{positive}}
\end{aligned}$$

which is strictly positive by Assumption (5). □

The intuition behind Proposition (5) comes from understanding the effects of two opposing forces in play. On the one hand, low-paying employers and specialized firms reduce job amenities, widening the gap with respect to high-paying employers and vertical firms. This effect across categories, which is the most intuitive, increases overall variance. Second, the variance within specialized employers and low-paying firms tends to shrink, as job amenities fall proportionally to the minimum wage change. This effect at the bottom of the distribution lowers overall variance, strengthening as the initial dispersion in job amenities among these firms increases.

Assumption (5) curbs the second force in play by bounding the initial dispersion in job amenities at the bottom. A first argument is that workers constrained by the minimum wage are unlikely to differ much from one another. In particular, because differences in productivity and outside options should have faded over time with worker flows. Second, after many years of wage-amenities tradeoffs in response to higher wage floors, the lower tail of the job amenities distribution should have compressed.

Nevertheless, note that the assumption is not a necessary condition and can be relaxed. In practice, the intuition suggests that the between-group effect is likely to dominate, leading to an overall increase in the variance of job amenities in response to minimum wage increases, even if within-group differences exert some non-negligible downward pressure.

Proposition 6 (Wage spillovers)**.** *Minimum wage increases propagate through the wage distribution by proportionally raising wages of high-paid workers specialized firms.*

Proof. The result follows immediately from Proposition (2), which shows the high-paid worker's wage increases linearly with $\underline{w}$. That is,

$$\frac{\partial \mathbb{E}_{\mathcal{S}}\left[w_H\right]}{\partial \underline{w}} > 0$$

□

From Proposition (6), the framework anticipates positive spillovers up to the percentile occupied by the highest-paid worker in specialized firms. Namely, employers with access to more affordable job amenities but limited workplace heterogeneity, or vice versa. More importantly, it highlights two features worth mentioning. First, spillovers are unrelated to other moments often used in the literature to interpret wage-floor effects, such as the median or average wage. Second, monotonicity in spillovers across the wage distribution may not necessarily hold. In particular, if the workplace heterogeneity of firms paying higher wage premiums on average exceeds that of firms with lower wage premiums. A similar argument can be made for the unitary price of job amenities.

B. Firm-specific wage premiums

The procedure for retrieving firm-level job amenities in Subsection III.B requires estimating firm-specific wage premiums within the SCC sample. In this section, I describe the empirical approach, report the estimation results, and present evidence addressing potential concerns about systematic bias.

I employ a model for wage determination (henceforth, the AKM model) based on Abowd et al. (1999), Card et al. (2013, 2018), Song et al. (2019), and Gerard et al. (2021), where the log of the hourly wage w_{it}, paid to worker i in quarter t, depends on observed and unobserved permanent worker and firm components, and time-varying observable characteristics

$$\log\ w_{it} = \alpha_i + \psi_{J(it)} + X_{it} \cdot b + \nu_{it}, \quad \nu \sim N(0, \sigma^2_\nu) \tag{B1}$$

where the worker fixed effect α_i (WFE) captures returns related to the inherent characteristics of individual i, such as schooling, skills, or cohort. The establishment fixed effect ψ_j (EFE) represents the wage premium that firm j pays to all employees, including profit-sharing, rents, or compensating differentials. b represents returns to time-varying worker characteristics, including an education-specific cubic polynomial in age and education-specific year fixed effects. [14] $J(it)$ is a mapping function that indicates the employer of worker i in quarter t. The residual term ν_{it} captures any other potential transitory fluctuations.

I estimate Equation (B1) in the AKM sample described in Subsection II.B, separately for the periods 2012-2015 and 2016-2019. As in the rest of the chapter, the analysis centers on workers aged 18 to 54 with open-ended urban contracts in the private sector. I next detail the data construction and cleaning procedure, designed to render the sample appropriate for the estimation of firm-specific firm-premiums.

AKM sample. I collect all employment spells to determine the modal legal status of firms, and the modal gender, nationality, and cohort of workers. I drop non-Brazilian

[14] Worker fixed effects and time-varying characteristics cannot be separately identified without normalization. I follow Card et al. (2018) and Gerard et al. (2021) by centering the education-specific polynomials on the last quarter of the sample, around 40 years old for male workers and around 35 years old for female workers. Furthermore, I define five identical education dummies: incomplete elementary school, complete elementary school, middle school, high school, and college.

workers, observations with missing values for gender, cohort, schooling, hours, or wages, and individuals with less than a year of potential experience. Thus, workers must be at least one year older than the theoretical graduation age for the highest degree they have attained.

I construct quarterly employment histories, determine transitions, and afterward assign workers with multiple employment relationships to the job with the highest quarterly earnings. That way, I can keep track of transitions in and out of the final sample separately from transitions in and out of non-employment. When the worker's identifier (*Cadastro de Pessoas Físicas*, CPF) appears in the *National Registry of Firms* (CNPJ, for its acronym in Portuguese), I can identify transitions to formal self-employment. Therefore, in my setting, non-employment consists only of unemployment and informal work. Finally, I remove the entire employment history of any individual whose log hourly wage changes by more than 100 log points from one quarter to the next.

The last two restrictions are tightly interconnected. First, firm-specific wage premiums are identified solely through worker flows, so I need to further restrict the sample to the largest weakly connected set of workers and firms. That is, the set of firms connected through at least one hire or separation. Second, I drop singletons (workers and firms I only observe once), which can introduce unnecessary noise into the analysis. Dropping singletons can create new partitions in the connected set, and removing observations from the connected component can, in turn, produce additional singletons. I therefore repeat this process iteratively until I remove no further employment spells.

1. Identifying conditions

The AKM framework relies on three identifying conditions. First, that worker and firm heterogeneity are roughly fixed over time. Second, that firms in the weakly connected set are sufficiently interconnected through worker mobility. Third, that worker mobility is uncorrelated with time-varying wage residuals in the estimation.

Fixed worker and firm heterogeneity. The AKM model assumes that worker and firm components of wage determination are time-invariant. Although it hardly holds in general, worker skills and firm pay policies tend to change gradually, making the restriction more plausible as the estimation window shortens. I estimate firm wage premiums over a four-year window, a relatively short panel compared to other studies.[15] A shorter horizon has the drawback of reducing the number of moves, raising concerns about limited mobility.

Limited mobility. Among sparsely linked weakly connected sets, estimates of firm-specific wage premiums become noisy. Although individual coefficients in Equation (B1) remain unbiased, noise leads to an overstatement of the variance of EFE at the detriment of the covariance between WFE and EFE. The literature has developed several methods to obtain unbiased variance estimates, such as shrinkage, leave-one-out estimators (Kline et al., 2020), and finite-mixture methods (Abowd et al., 2019; Bonhomme et al., 2019; Lentz et al., 2023). Given that my interest centers on individual firm-specific fixed effects, which remain unbiased, I do not implement any of these workarounds.

Nonetheless, a second drawback of limited mobility concerns representativeness. A shorter horizon reduces the likelihood that smaller firms will hire from or employ workers moving to another firm in the weakly connected set. I take advantage of Brazil's high turnover among smaller firms, with most moves occurring within a single year. Thus, I construct quarterly rather than annual employment histories, which allows me

[15] Five years in Engbom and Moser (2022), six in Card et al. (2013) and Sorkin (2018), seven in Song et al. (2019), eight years in Card et al. (2016), and thirteen in Gerard et al. (2021).

to stick to the standard assumption in the literature of keeping a unique employment spell per individual per period (e.g., Card et al., 2013, 2016, 2018; Gerard et al., 2021).

Exogenous mobility. The AKM framework obtains biased estimates when moves systematically depend on other components beyond worker and firm fixed effects or the included time-varying controls. If that were the case, worker mobility would correlate with time-varying match-specific shocks, the worker's employment history, or other transitory wage components, so that the specification would suffer from omitted-variable bias.

Nonetheless, estimates can be unbiased even when individual workers respond to positive or negative shocks at their current jobs, or if wage changes around job transitions combine true firm-specific wage premiums with match-specific gains or losses included in ν_{it}. As long as these deviations average out across movers, estimated firm effects will capture pay premia differentials rather than sorting. Therefore, the exogenous mobility assumption requires that, conditional on individual and firm characteristics, mobility is as good as random.

Card et al. (2013) developed a series of checks that would flag clear violations of the exogenous mobility condition. Gerard et al. (2021) run this analysis using RAIS data from 2002 to 2014 and confirm that wage changes for movers in Brazil are broadly consistent with the assumption. Next, I replicate the analysis using quarterly employment spells between 2012 and 2019 and arrive at the same conclusion.

A first approach proposes fully saturating the AKM framework by including a match-specific component, m_{ij}. If match effects were essential in wage determination, this version of the model should display a significant improvement in fit. Table (B1) shows that the non-saturated version already reports an adjusted R^2 above 0.95 across samples; I therefore skip this step.

The second test follows from Equation (B1). If the AKM model is correctly specified, then the following regression should satisfy an identity up to sampling error

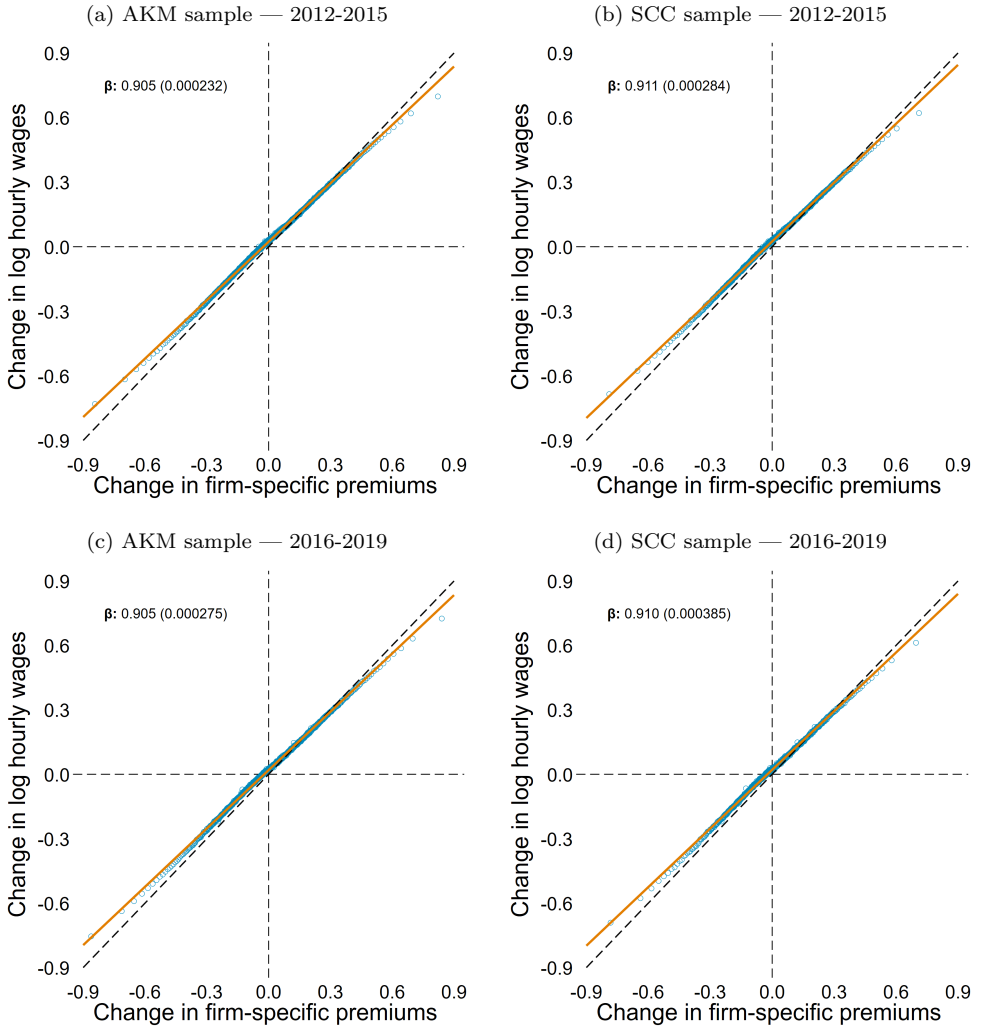
$$\mathbb{E}\left[w^r_{it+1} - w^r_{it}\right] = \psi_{J(it+1)} - \psi_{J(it)} + \mathbb{E}\left[\nu_{it+1} - \nu_{it}\right]$$

where $w^r_{it} = w_{it} - X_{it}\hat{b}$. Given that I focus on movers, t corresponds to the last quarter at the predecessor firm, and $t+1$ corresponds to the first quarter at the successor firm. Figure (B1) plots the binscatter for this specification, separately for the AKM and the SCC samples, and the periods 2012-2015 and 2016-2019. Notice the relationship does not follow an exact identity, though the linear fit across individual transitions reports a slope above 0.9 in all cases. Therefore, one can conclude that firm-specific wage premiums likely contain some noise but do not immediately reveal large biases.

Finally, if systematic wage gains or losses drove worker mobility, one would expect at least one of two patterns. First, workers moving up and down the job ladder between the same two firms would exhibit wage changes of different magnitudes. Second, wage trajectories would display pre-trends at the predecessor firm or post-trends at the successor firm. Figure (B2) plots the canonical event study, which compares wage histories among workers moving across firms in different wage premium quartiles.

I do not restrict the analysis to job-to-job transitions, but focus on workers who stayed at the same firm for at least one year before and after moving. Thus, t_{OLD} denotes the last quarter at the predecessor firm, while $t_{OLD} - 1$ corresponds to four quarters before. Likewise, t_{NEW} denotes the first quarter at the successor firm, while $t_{NEW} + 1$ corresponds to four quarters after. The plot displays flat trends before and after the job transitions. In particular, across establishments within the same quartile, differences in log hourly wages are relatively small, suggesting no broad mobility pre-

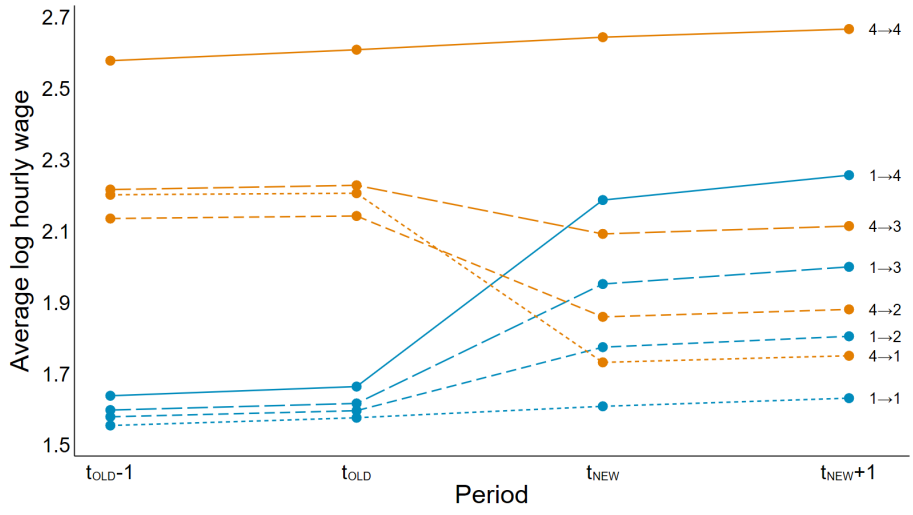
FIGURE B1 — WAGE AND PREMIUMS CHANGES FOR MOVERS



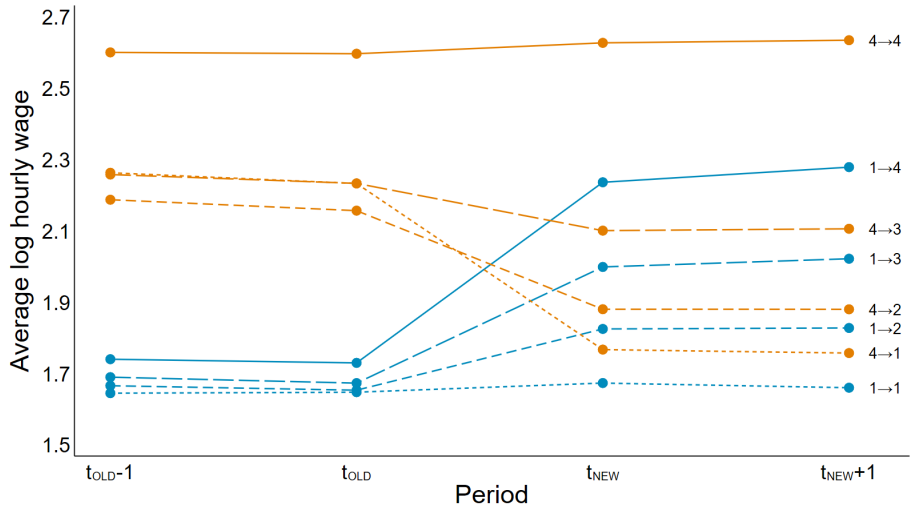
Notes: The figure plots changes in log hourly wages against changes in firm-specific wage premiums among movers. Log hourly wages are residualized with respect to time-varying characteristics using estimates from Equation (B1) and correspond to hourly rates in the last quarter at the predecessor firm and in the first quarter at the successor firm. Panel (a) and (c) use all transitions among firms in the AKM sample. Panel (b) and (d) use all transitions among firms in the SCC sample. Blue dots correspond to a binscatter regression following the method in Cattaneo et al. (2025). The black-dashed diagonal corresponds to the 45-degree line. The solid orange line plots the best linear fit. The corresponding coefficient and standard error (95% confidence) are reported at the upper left corner. Estimations are based on quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector.

FIGURE B2 — EVENT STUDIES OF WAGE TRAJECTORIES

(a) 2012–2015



(b) 2016–2019



Notes: The figure plots wage trajectories for workers changing jobs from a predecessor firm in the bottom or top quartile to a successor firm in any of the other quartiles. I use quarterly RAIS employment spells in the AKM sample for workers aged 18-54 with open-ended contracts in the urban private sector. The analysis focuses on workers who stayed at the same firm for at least one year before and after changing jobs, although not necessarily to job-to-job transitions. Thus, t_{OLD} denotes the last quarter at the predecessor firm, while $t_{OLD} - 1$ corresponds to four quarters before. Likewise, t_{NEW} denotes the first quarter at the successor firm, while $t_{NEW} + 1$ corresponds to four quarters after. I define quartiles using estimates for firm fixed effects from Equation (B1) and calculate average log hourly wages for each type of transition over time. Panel (a) performs the analysis for the period 2012-2015, while Panel (b) replicates the exercise for the period 2016-2019. Blue lines correspond to predecessors in the bottom quartile, while orange lines correspond to predecessors in the top quartile. The shorter the dash, the lower the quartile of the successor firm. For clarity, I report the quartiles next to each line for both the predecessor and successor firms.

miums. Consistently, worker flows across establishments in different quartiles showcase roughly symmetric wage differentials.

2. Estimation results

The following subsection presents the estimation results for the AKM framework. Table (B1) performs the canonical variance decomposition in Abowd et al. (1999). Namely,

$$\mathbb{V}[w_{it}] = \underbrace{\mathbb{V}[\alpha_i] + \mathbb{V}[X_{it}\,b] + 2\,\mathbb{C}ov[\alpha_i, X_{it}\,b]}_{\substack{\text{worker time-invariant and} \\ \text{time-varying characteristics}}} + \underbrace{\mathbb{V}[\psi_j]}_{\substack{\text{firm} \\ \text{wage premia}}} + \underbrace{2\Big(\mathbb{C}ov[\alpha_i, \psi_j] + \mathbb{C}ov[\psi_j, X_{it}\,b]\Big)}_{\text{worker sorting}} + \mathbb{V}[\nu_{it}]$$

These estimates quantify the relative importance of worker- and firm-specific components on wage determination. Notice, however, that as I mentioned before, the variance of firm pay premia is upward-biased at the detriment of $\mathbb{C}ov[\alpha_i, \psi_j]$.

As mentioned above, using quarterly employment histories in Brazil increases the number of movers across samples, with a mover-to-worker ratio close to 1. Although the ratio decreases for the SCC samples, it remains relatively high compared to similar estimations, ranging from 0.48 to 0.70. [16]

EFE explains less of the variance in the SCC sample. Two immediate potential explanations arise. On the one hand, establishments in the SCC tend to be larger and may exhibit a more similar distribution of pay premiums. On the other hand, these employers are more interconnected through worker flows. Remember that, by construction, firms must have at least one hire and one separation that connects them with other employers in the SCC. As a result, firm fixed effects in the SCC may contain less noise and be more precisely estimated.

Table (B2) turns attention to an alternative variance decomposition following Song et al. (2019). Comparing results across SCC samples shows that within-firm wage inequality remained largely stable over time, with differences driven almost entirely by changes in the residual component. In contrast, most of the decline in wage inequality in Brazil over time is explained by determinants of the between-firm wage variance, including declines in worker segregation, worker sorting, and firm-specific pay premia.

$$\mathbb{V}[w_{it}] = \overbrace{\underbrace{\mathbb{V}[\overline{\alpha}_j] + \mathbb{V}\left[\overline{X}_j\,b\right] + 2\,\mathbb{C}ov\left[\psi_j, \overline{X}_j\,b\right]}_{\substack{\text{worker} \\ \text{segregation}}} + \underbrace{\mathbb{V}[\psi_j]}_{\substack{\text{firm} \\ \text{wage premia}}} + \underbrace{2\Big(\mathbb{C}ov[\overline{\alpha}_j, \psi_j] + \mathbb{C}ov\left[\overline{\alpha}_j, \overline{X}_j\,b\right]\Big)}_{\text{worker sorting}}}^{\textbf{Between-firm component}} + \overbrace{\mathbb{V}[\alpha_i - \overline{\alpha}_j] + \mathbb{V}\left[X_{it}\,b - \overline{X}_j\,b\right] + 2\,\mathbb{C}ov\left[\alpha_i - \overline{\alpha}_j, X_{it}\,b - \overline{X}_j\,b\right] + \mathbb{V}[\nu_{it}]}^{\textbf{Within-firm component}}$$

[16] For instance, Gerard et al. (2021), who explicitly report the number of movers in their samples, have a ratio of around 0.15.

Table B1 —Variance decomposition of log hourly wages

	2012-2015				2016-2019			
	AKM		SCC		AKM		SCC	
$\mathbb{E}(w)$	2.108		2.188		2.155		2.260	
$\mathbb{V}(w)$	0.396		0.432		0.353		0.401	
$\mathbb{V}(\alpha)$	0.269	67.78%	0.302	69.77%	0.246	69.82%	0.289	71.99%
$\mathbb{V}(\psi)$	0.044	11.19%	0.037	8.49%	0.042	11.95%	0.032	8.04%
$\mathbb{V}(X\,b)$	0.005	1.37%	0.006	1.29%	0.005	1.33%	0.005	1.21%
$\mathbb{V}(\nu)$	0.015	3.91%	0.017	3.94%	0.012	3.37%	0.013	3.31%
$2\cdot\mathbb{C}\text{ov}(\alpha,\psi)$	0.070	17.63%	0.078	18.15%	0.055	15.44%	0.069	17.13%
$2\cdot\mathbb{C}\text{ov}(\alpha,X\,b)$	-0.007	-1.77%	-0.007	-1.67%	-0.007	-1.90%	-0.007	-1.81%
$2\cdot\mathbb{C}\text{ov}(\psi,X\,b)$	-0.000	-0.11%	-0.000	-0.10%	0.000	0.00%	-0.000	-0.00%
$\mathbb{C}\text{ov}(w,\psi)$	0.079	19.95%	0.076	17.51%	0.069	19.67%	0.067	16.60%
Establishments	3,323,529		765,419		3,126,974		498,859	
Workers	49,079,671		39,338,017		44,137,460		30,704,508	
Moves	49,198,880		27,714,180		40,228,874		14,821,121	
Employment spells	497,308,284		372,771,472		451,144,482		291,773,507	
R^2	0.9563				0.9624			

Notes: The table decomposes wage variation in Brazil, separately for the periods 2012-2015 and 2016-2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. The AKM sample centers on employers in the weekly connected set of movers. The SCC focuses on employers in the strongly connected set of actively recruited employees. The analysis follows the estimation of Equation (B1) in the AKM sample and decomposes the variation in log hourly wages (w) into worker fixed effects (α), firm fixed effects (ψ), time-varying controls ($X\,b$, including education-specific cubic polynomials of age and year fixed effects), residuals (ν), and the corresponding covariances. $\mathbb{C}ov(w,\psi)$ reports the sum of all firm-related components, equal to $\mathbb{V}(\psi)+\mathbb{C}ov(\psi,\alpha)+\mathbb{C}ov(\psi,X\,b)$. The last five rows report the numbers of unique establishments, unique workers, moves, employment spells (worker-year observations), and R-squared of the estimation. The SCC columns use the same estimation but replicate the analysis only for worker-year observations in the SCC sample.

Table B2 —Variance decomposition between and within establishments

	2012-2015				2016-2019			
	AKM		SCC		AKM		SCC	
$\mathbb{E}(w)$	2.108		2.188		2.155		2.260	
$\mathbb{V}(w)$	0.396		0.432		0.353		0.401	
Between-firm variance	0.229	57.71%	0.237	54.87%	0.199	56.30%	0.212	52.79%
$\mathbb{V}(\overline{\alpha})$	0.118	29.85%	0.126	29.19%	0.105	29.71%	0.114	28.49%
$\mathbb{V}(\psi)$	0.044	11.19%	0.037	8.49%	0.042	11.95%	0.032	8.04%
$\mathbb{V}(\overline{X}\,b)$	0.001	0.16%	0.001	0.12%	0.001	0.19%	0.001	0.14%
$2\cdot\mathbb{C}\text{ov}(\overline{\alpha},\psi)$	0.070	17.63%	0.078	18.15%	0.055	15.44%	0.069	17.13%
$2\cdot\mathbb{C}\text{ov}(\overline{\alpha},\overline{X}\,b)$	-0.004	-1.01%	-0.004	-0.98%	-0.004	-0.99%	-0.004	-1.00%
$2\cdot\mathbb{C}\text{ov}(\psi,\overline{X}\,b)$	-0.000	-0.11%	-0.000	-0.10%	0.000	0.00%	-0.000	-0.00%
Within-firm variance	0.168	42.29%	0.194	44.99%	0.154	43.70%	0.189	47.07%
$\mathbb{V}(\alpha-\overline{\alpha})$	0.150	37.93%	0.175	40.57%	0.142	40.11%	0.175	43.50%
$\mathbb{V}(X\,b-\overline{X}\,b)$	0.005	1.21%	0.005	1.17%	0.004	1.14%	0.004	1.08%
$\mathbb{V}(\nu)$	0.015	3.91%	0.017	3.94%	0.012	3.37%	0.013	3.31%
$2\cdot\mathbb{C}\text{ov}(\alpha-\overline{\alpha},X\,b-\overline{X}\,b)$	-0.003	-0.76%	-0.003	-0.69%	-0.003	-0.91%	-0.003	-0.82%

Notes: The table decomposes wage variation within and across firms in Brazil, separately for the periods 2012-2015 and 2016-2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. The AKM sample centers on employers in the weekly connected set of movers. The SCC focuses on employers in the strongly connected set of actively recruited employees. The analysis follows the estimation of Equation (B1) in the AKM sample and decomposes the variation in log hourly wages (w) into between- and within-firm components, as in Song et al. (2019). The between-firm component includes firm fixed effects (ψ), firm-level averages of worker fixed effects ($\overline{\alpha}$), and time-varying controls ($\overline{X}\,b$, including education-specific cubic polynomials of age and year fixed effects), along with their corresponding covariances. The within-firm component includes idiosyncratic variation from worker fixed effects ($\alpha-\overline{\alpha}$) and time-varying controls ($X\,b-\overline{X}\,b$) within firms, residuals (ν), and their covariance.

C. Firm-specific job amenities

I estimate firm-specific job amenities following Sorkin (2018) and Lehmann (2025). The underlying job search model shares the assumptions listed in Section I, but introduces two further ones. First, employed and non-employed workers receive the same number of offers from an identical distribution, making the job search non-directed and effortless. Second, workers value firms identically, up to i.i.d. idiosyncratic preferences, which yields a unique firm ranking—or job ladder—across all workers.

Employers offer contracts $\langle w_{it}, a_j \rangle$ and differ in size g_j, exogenous separation rates δ_j, exogenous reallocation rates ρ_j, and exogenous hiring rates f_j.

A worker i values employment at firm j according to the flow payoff $v_j(w_{it}, a_j)$ and the associated continuation value, discounted at rate β. With probability δ_j, the firm lays off the worker. Conditional on layoff, the worker reallocates to a new job before the end of the next quarter with probability ρ_j; otherwise, she becomes non-employed. In the absence of a separation, the worker receives an offer from firm k with probability λ_E and decides whether to accept it and quit or remain at her current firm. With probability $1-\lambda_E$, she receives no offer but may still choose to exit to nonemployment. The following specification synthesizes the associated Bellman equation

$$
\begin{aligned}
\underbrace{V_j^e(w_{it}, a_{it})}_{\text{value of being at j}} &= \underbrace{v_j(w_{it}, a_{it})}_{\substack{\text{flow}\\ \text{payoff}}} + \beta \cdot \mathbb{E}\Bigg[\underbrace{\delta_j(1-\rho_j) \int_{\iota_1} \{V^n + \iota_1\}\, dI}_{\substack{\text{exogenous}\\ \text{job destruction}}} + \underbrace{\delta_j \rho_j \sum_k \int_{\iota_2} \{V_k^e + \iota_2\}\, dI\, \tilde{f}_k}_{\text{exogenous employer-to-employer}} \\
&+ \underbrace{(1-\delta_j)}_{\substack{\text{no exogenous}\\ \text{job destruction}}} \times \Bigg(\underbrace{\overbrace{\lambda_E}^{\text{offer}} \sum_k \int_{\iota_3} \int_{\iota_4} \max\{\overbrace{V_k^e + \iota_3}^{\text{accept}}, \overbrace{V_j^e + \iota_4}^{\text{reject}}\}\, dI\, dI\, f_k}_{\text{endogenous employer to employer}} \\
&+ \underbrace{\overbrace{(1-\lambda_E)}^{\text{no offer}} \int_{\iota_5} \int_{\iota_6} \max\{\overbrace{V^n + \iota_5}^{\text{leave}}, \overbrace{V_j^e + \iota_6}^{\text{stay}}\}\, dI\, dI\, f_k}_{\text{endogenous job destruction}} \Bigg)
\end{aligned}
$$

The empirical strategy revolves around workers who receive offers from actively recruiting firms. The decision to remain with the incumbent firm or to transition to a new job is what reveals their preferences for employers. Given that the framework assumes that all workers broadly agree on the firm ranking, relative flows across any pair of firms help determine the relative value between the two employers.

$$
\underbrace{\frac{M_{jk}}{M_{kj}}}_{\substack{\text{relative}\\ \text{flows}}} = \underbrace{\frac{f_j}{f_k}}_{\substack{\text{relative}\\ \text{offers}}} \underbrace{\frac{g_k(1-\delta_k)}{g_j(1-\delta_j)}}_{\substack{\text{effective relative}\\ \text{firm size}}} \underbrace{\frac{\mathbb{P}(j \succ k)}{\mathbb{P}(k \succ j)}}_{\substack{\text{relative}\\ \text{firm values}}}
$$

where M_{kj} represents employer-to-employer transitions from firm j to employer k.

Sorkin (2018) demonstrates that flows across every pair of firms are not necessary. As long as one focuses on firms hiring workers from or employing workers transitioning to other employers in the SCC, one can identify relative firm values from the entire worker flow network.

$$
\frac{\overbrace{\sum_{j \in J} M_{kj} \exp\left(V_j^e\right)}^{\text{value-weighted hires}}}{\underbrace{\sum_{j \in J} M_{jk}}_{\text{quits}}} = \underbrace{\exp\left(V_k^e\right)}_{\text{firm value}}
$$

Lehmann (2025) builds on Sorkin (2018) by imposing a functional form on workers' indirect utility (not the flow payoff)

$$V_{it}^{j} = V_{j}^{e}\left(w_{it}, a_{it}\right) = w_{it}^{\gamma} \cdot a_{it}$$

where γ represents the elasticity of substitution between wages and job amenities.

Next, I specify that a_{it} depends on firm-level job amenities $\chi_{J(it)}$ and idiosyncratic preferences ι_{it},

$$\log \;\; a_{it} = \chi_{J(it)} + \iota_{it}, \qquad \iota \sim N(0, \sigma_{\iota}^{2})$$

and wages are jointly determined by worker and firm characteristics, following the AKM model specification in Section B in the Appendix

$$\log \;\; w_{it} = \alpha_i + \psi_{J(it)} + X_{it} \cdot b + \nu_{it}, \qquad \nu \sim N(0, \sigma_{\nu}^{2})$$

Thus, I can rewrite each worker's indirect utility as

$$\log \;\; V_{it}^{j} = \gamma\left(\alpha_i + \psi_j + X_{it}\beta\right) + \chi_j + \underbrace{\gamma\nu_{it} + \iota_{it}}_{\varepsilon}, \qquad \varepsilon_{it} \sim N(0, \underbrace{\gamma^2\sigma_{\nu}^2 + \sigma_{\iota}^2}_{\sigma_{\varepsilon}^2})$$

where ε are idiosyncratic and normally distributed i.i.d. shocks. Notice that ε combines individual deviations around common preferences for firm-specific amenities with temporary fluctuations in wages. The specification runs along the same lines as Lentz et al. (2023), who assume that workers sort themselves across firms according to underlying wage distributions, draw an initial wage from those distributions when they transition to a new job, and receive new realizations that depend on past wages. Moreover, since estimates of firm-specific wage premiums ψ_j can be noisy, epsilon also captures noise from the AKM estimation.

A worker i, who receives an offer from firm k while employed at firm j, compares the expected utilities V_{it}^{j} and V_{it}^{k}. Whenever the worker chooses to transition to the new job, it reveals that she ranks compensation in firm k higher than in firm j on the job ladder. That is,

$$\begin{aligned}
\mathbb{P}(k \succ j) &= \mathbb{P}\left(\log V_{it}^{k} > \log V_{it}^{j}\right) \\
&= \mathbb{P}\Big(\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j) > (\varepsilon_{1it} - \varepsilon_{2it})\Big) \\
&= \Phi\left(\frac{\gamma\big(\psi_k - \psi_j\big) + \big(\chi_k - \chi_j\big)}{2\sigma_{\varepsilon}}\right)
\end{aligned}$$

Therefore, for a worker that values compensation at two firms equally, $V_{it}^{j} = V_{it}^{k}$, any difference in firm-specific pay premiums, $\psi_j - \psi_k$, reveal compensating differentials due to uneven job amenities, $\chi_j - \chi_k$, or idiosyncratic fluctuations in preferences and wages, $\varepsilon_{1it} - \varepsilon_{2it}$.

I am hereby deviating from the original approach in Lehmann (2025) for two reasons. First, instead of using wages directly, I employ Equation (B1) from the AKM model, which accounts for wage differentials originating in time-varying worker characteristics and temporary wage fluctuations. Second, I do not impose $\sigma_{\varepsilon} = 1/2$, which undermines my ability to interpret γ estimates but reveals the importance of appropriately scaling job amenities χ_j.

1. Estimation

I estimate the model by maximum likelihood separately for the periods 2012-2015 and 2016-2019. In particular, I solely use endogenous employer-to-employer transitions among firms in the SCC sample, using the following likelihood function

$$\mathcal{L}(\theta|X) = \prod_{n=1}^{N} \Phi\left(\frac{\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j)}{2\sigma_\varepsilon}\right)^{\frac{1}{g_j}\frac{1}{f_k}} \tag{C1}$$

where g_j and f_k weight individual contributions by the size of the predecessor firm j and the number of offers made by the successor firm k. The vector θ of parameters has size $J+1$: one for the elasticity of substitution *gamma* and J for each of the firm-specific job amenities χ_j. To estimate the model, I randomly set $M = 10$ initial values for θ, which I maintain across all jackknife and bootstrap estimations for consistency. For each estimation, I keep the parameter set that maximizes the likelihood function.

Notice, however, that parameters are identified up to a scaling parameter $2\sigma_\varepsilon$, such that

$$\hat{\gamma} = \frac{\gamma}{2\sigma_\varepsilon} \qquad \text{and} \qquad \hat{\chi}_j = \frac{\chi_j}{2\sigma_\varepsilon} \; \forall j \in J$$

which requires me to follow a series of steps described in Lehmann (2025) before computing the variance of job amenities across samples.

Incidental parameter bias. Since the estimator is nonlinear, and the number of establishments has a similar order of magnitude as the number of transitions in the sample, $\hat{\gamma}$ suffers from incidental parameter bias. Therefore, I implement the jackknife correction detailed in Lehmann (2025), inspired by the original proposal of Hahn and Newey (2004).

Let N be the number of transitions in the sample. Each establishment uses, on average, $T = {}^{2N}/_{J}$ observations to identify the corresponding fixed effect, which could be interpreted, analogously to the panel data literature, as the length of the time series. Similar to the original approach, the procedure randomly splits transitions into T separate groups. However, instead of discarding the transitions in the corresponding group for each jackknife replication, this technique weights down the individual contributions by a factor of 1 to 1000. Otherwise, randomly discarding transitions could break the sample's strongly connected structure that is necessary for identification.

Each jackknife replication yields a new estimate $\hat{\gamma}^{T-1}$ of the elasticity of substitution, reducing the bias from $\mathcal{O}(1/T)$ to $\mathcal{O}(1/T^2)$ by using the following equation

$$\hat{\gamma}_{JK} = T\,\hat{\gamma}^{T} - (T-1)\,\mathbb{E}\left[\hat{\gamma}^{T-1}\right]$$

where $\hat{\gamma}^{T}$ corresponds to the baseline estimate.

Job amenities scaling. As mentioned before, estimates for $\hat{\gamma}_{JK}$ and $\hat{\chi}_j$ are not identified separately from the scaling parameter $2\sigma_\varepsilon$. However, the ratio between the two can be established, irrespective of the scaling parameter. In other words, I can define the following,

$$\hat{\chi}^{BRL} = \frac{\hat{\chi}}{\hat{\gamma}_{JK}} = \frac{\chi}{\gamma_{JK}}$$

which corresponds to the value of firm-specific job amenities in log-BRL (*reais brasileiros*). Therefore, hereafter, I refer to job amenities in this unit of measure.

Variance correction. As in the AKM framework, firm-specific job amenities are estimated with noise, which leads to an upward bias in the plugin variance estimate.

Therefore, I implement the correction discussed in Lehmann (2025).

The first step is to estimate the baseline noise. I fix $\gamma = \hat{\gamma}_{JK}$ and bootstrap the likelihood function $B = 50$ times. To maintain the same network structure in the SCC, instead of randomly sampling transitions, I randomly assign them positive weights ω_i, such that

$$\sum_N \omega_n = N$$

Next, I normalize each estimation so that the average level of job amenities across all establishments equals zero, and define a firm-specific noise term π_j^2, which corresponds to the variance in firm-specific job amenities across all bootstrap samples.

The second step is to recalculate the plugin estimate of the variance using

$$\mathbb{V}\left[\chi^{BRL}\right] = \frac{\sum_J size_j \left[\left(\frac{J}{J-1}\right)\left(\hat{\chi}^{BRL} - \overline{\chi}^{BRL}\right)^2 - \pi_j^2\right]}{\sum_J size_j}$$

where $size_j$ denotes the size of firm j and $\overline{\chi}^{BRL}$ represents the employment weighted average of job amenities.

2. *Estimation results*

Table (C1) presents key information on the maximum likelihood estimation using Equation (C1), separately for the periods 2012-2015 and 2016-2019. The first thing worth noting is the sharp decline in endogenous employer-to-employer transitions in Brazil over time. The immediate consequence is that job amenities in the latest period are more noisily estimated. To account for this, I construct 95% confidence intervals by reporting the 5th and 95th percentiles of all corrected plugin variance estimations from each bootstrap.

The confidence intervals across samples confirm the suspicion that estimates in the later period contain more noise, yet still suggest a significant difference in magnitude.

Table C1 —MLE estimation of firm-specific job amenities in log-BRL

2012-2015				2016-2019			
Baseline estimation							
M	10	**Transitions** (N)	8,002,800	**M**	10	**Transitions** (N)	4,609,276
T	21	**Firms** (J)	765,419	**T**	19	**Firms** (J)	498,859
γ	2.938	**Bias:** $O(1/T)$	0.0476	γ	3.855	**Bias:** $O(1/T)$	0.0526
Jackknife correction							
γ_{JK}	5.412	**Bias:** $O(1/T^2)$	0.0022	γ_{JK}	2.680	**Bias:** $O(1/T^2)$	0.0027
Variance of job amenities							
B	50	**Non-corrected**	0.0632	**B**	50	**Non-corrected**	0.4046
		Corrected	0.0627 (0.0401,0.0891)			**Corrected**	0.4052 (0.1344,0.6850)

Notes: The table presents estimates, corrections, and key information on the maximum likelihood estimation using Equation (C1), separately for the periods 2012-2015 and 2016-2019. The sample consists solely of quarterly employer-to-employer transitions between establishments in the SCC sample. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. I randomly set $M = 10$ initial values for the parameter vector θ, which I keep across all T jackknife and $B = 50$ bootstrap estimations for consistency. T corresponds to the average number of transitions used to identify each individual job amenities fixed effect $2N/J$. For each sample, I report the baseline γ and the jackknife correction γ_{JK}, along with their respective biases. Finally, I report the baseline and corrected plugin estimators for the variance of job amenities in log-BRL.